

BEACON EDUCATIONAL SERIES

Mathematics for Basic 9

A Comprehensive Question Bank
with Marking Schemes

Curriculum-Aligned Practice Papers
for Basic 9 (JHS 3) Students and Teachers

Beacon Educational Consult
2026 Edition

Mathematics for Basic 9
First Edition, 2026

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This book is designed to support the Ghana Education Service (GES) curriculum for Junior High School. It is intended as a supplementary resource for teachers and students.

The questions in this book are original practice materials created for educational purposes. They are not official BECE or GES examination questions.

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PREFACE

This book has been developed to support Junior High School Basic 9 (JHS 3) students and teachers in their study of Mathematics, as prescribed by the Ghana Education Service (GES) and the National Council for Curriculum and Assessment (NaCCA).

The book contains a comprehensive collection of practice questions organised according to the curriculum's strand and sub-strand structure. Each paper includes both multiple-choice questions (Section A) and structured theory questions (Section B), reflecting the format of BECE examinations.

Key Features

- Curriculum-Aligned Content — All questions mapped to specific content standards.
- Comprehensive Coverage — Topic-based papers covering all major strands.
- Detailed Marking Schemes — Complete answer keys with method marks and marking guidance.
- Professional Diagrams — Clear, scientifically accurate illustrations.
- Cognitive Demand — Questions spanning all levels of Bloom's taxonomy.

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ANSWER KEY

Chapter 1: Ch10

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Solve $x + 3 > 5$.

- A. $x < 2$
- B. $x > 2$
- C. $x > 8$
- D. $x > -2$

2. Solve $2x \leq 8$.

- A. $x \leq 16$
- B. $x < 4$
- C. $x \leq 4$
- D. $x \geq 4$

3. Solve $x - 4 < 1$.

- A. $x < 5$
- B. $x < 4$
- C. $x > 5$
- D. $x < -3$

4. Solve $3x > 12$.

- A. $x < 4$
- B. $x > 9$
- C. $x > 36$
- D. $x > 4$

5. Solve $-x < 3$.

- A. $x > -3$
- B. $x > 3$
- C. $x < 3$
- D. $x < -3$

6. Solve $-2x \geq 6$.

- A. $x \leq 3$
- B. $x \leq -3$
- C. $x \geq -3$
- D. $x \geq 3$

7. Solve $x + \frac{1}{2} \geq -\frac{3}{2}$.

- A. $x \leq -2$
- B. $x \geq -1$
- C. $x \geq -2$
- D. $x \geq 2$

8. Solve $3x - 2 \geq 7$.

- A. $x \geq 3$
- B. $x \geq 5$
- C. $x \geq 2$
- D. $x \leq 3$

9. Solve $5 - x > 2$.

- A. $x > -3$
- B. $x < 3$
- C. $x > 3$
- D. $x < -3$

10. Solve $x/3 < 2$.

- A. $x < 6$
- B. $x > 6$
- C. $x < 2/3$
- D. $x < 3/2$

11. Solve $2x/3 \geq 4$.

- A. $x \geq 2$
- B. $x < 6$
- C. $x \geq 6$
- D. $x \leq 6$

12. Solve $4 - 3x \leq 10$.

- A. $x \leq 2$
- B. $x \leq -2$
- C. $x \geq 2$
- D. $x \geq -2$

13. Solve $5x + 2 < 3x + 10$.

- A. $x > 4$
- B. $x < 4$
- C. $x < 6$
- D. $x > 8$

14. Solve $3(x - 1) \geq 2x + 1$.

- A. $x \geq 4$
- B. $x \geq 2$
- C. $x \geq -4$
- D. $x \leq 4$

15. Solve $-(2/3)x + 3 \geq 3$.

- A. $x \geq 3$
- B. $x \geq 0$
- C. $x \leq 0$
- D. $x \leq 3$

16. Solve $\frac{1}{2}(5x - 5) < x + 5$.

- A. $x < 5$
- B. $x > 5$
- C. $x < 15$
- D. $x < 10$

17. The inequality shown on the number line is

- A. $x < 3$
- B. $x \leq 3$
- C. $x \geq 3$
- D. $x > 3$

18. The inequality shown on the number line is

- A. $x \leq 2$
- B. $x < -2$
- C. $x \leq -2$
- D. $x \geq -2$

19. An open (unshaded) circle on a number-line graph shows that

- A. the end value is included in the solution
- B. the solution is zero
- C. the arrow points left
- D. the end value is NOT included in the solution

20. A closed (shaded) circle on a number-line graph shows that

- A. there is no solution
- B. all numbers solve it
- C. the end value is included in the solution
- D. the end value is not included

- 21.** The inequality shown on the number line is
- A. $x > 4$
 - B. $x < 4$
 - C. $x \geq 4$
 - D. $x \leq 4$
- 22.** Which graph represents $x < 5$?
- A. open circle at 5 with arrow to the right
 - B. open circle at 5 with arrow to the left
 - C. closed circle at 5 with arrow to the left
 - D. closed circle at 5 with arrow to the right
- 23.** The solution of $-3x - 8 > -26$ is (curriculum exemplar)
- A. $x < -6$
 - B. $x > -6$
 - C. $x < 6$
 - D. $x > 6$
- 24.** The solution of $2x - 3 \leq 19$ is (curriculum exemplar)
- A. $x \leq 11$
 - B. $x < 11$
 - C. $x \geq 11$
 - D. $x \leq 22$
- 25.** The solution of $2x \geq 8$ is (curriculum exemplar)
- A. $x \leq 4$
 - B. $x > 4$
 - C. $x \geq 16$
 - D. $x \geq 4$
- 26.** The solution of $5y + 3 < 6 + 2y$ is (curriculum exemplar)
- A. $y < 1$
 - B. $y < 3$
 - C. $y > 3$
 - D. $y > 1$
- 27.** The solution of $7x/3 < 7$ is (curriculum exemplar)
- A. $x < 21$
 - B. $x < 3$
 - C. $x > 3$
 - D. $x < 1$

- 28.** The solution set $x < 6$ is illustrated on the number line with
- A. an open circle at 6 and an arrow to the left
 - B. an open circle at 6 and an arrow to the right
 - C. a closed circle at 6 and an arrow to the left
 - D. a closed circle at -6 and an arrow to the left
- 29.** The statement 'x is at least 7' is written as
- A. $x \leq 7$
 - B. $x < 7$
 - C. $x > 7$
 - D. $x \geq 7$
- 30.** The statement 'x is at most 7' is written as
- A. $x > 7$
 - B. $x \leq 7$
 - C. $x < 7$
 - D. $x \geq 7$
- 31.** Ama has GH¢50. She buys books at GH¢5 each and wants to keep at least GH¢20. The number n of books she can buy satisfies
- A. $n \geq 6$
 - B. $n \leq 10$
 - C. $n < 6$
 - D. $n \leq 6$
- 32.** Twice a number, increased by 5, is greater than 17. The number satisfies
- A. $x > 12$
 - B. $x > 6$
 - C. $x > 11$
 - D. $x < 6$
- 33.** Kofi's age a is at least 12 years. This is written as
- A. $a < 12$
 - B. $a > 12$
 - C. $a \geq 12$
 - D. $a \leq 12$
- 34.** A taxi charges GH¢3 per trip plus GH¢2 per kilometre. With at most GH¢23, the number of kilometres k satisfies
- A. $k \leq 10$
 - B. $k \geq 10$
 - C. $k \leq 13$
 - D. $k \leq 20$

- 35.** To pass, a student needs at least 45 marks. She has 30 marks so far. The mark t she must score in the final test satisfies
- A. $t > 75$
 - B. $t \geq 45$
 - C. $t \leq 15$
 - D. $t \geq 15$
- 36.** A rectangle has width 4 cm and a perimeter of at most 24 cm. Its length l satisfies
- A. $l \leq 20$
 - B. $l \leq 8$
 - C. $l \geq 8$
 - D. $l \leq 10$
- 37.** The sum of three consecutive integers is less than 30. The largest possible value of the smallest integer is
- A. 9
 - B. 10
 - C. 8
 - D. 27
- 38.** The solution of $\frac{1}{2}(2x + 3) \leq x + 2$ is
- A. no solution
 - B. $x \geq 2$
 - C. $x \leq 2$
 - D. all real numbers (always true)
- 39.** The values of x that satisfy both $x > 2$ and $x < 5$ are
- A. $x > 2$
 - B. $x < 2$
 - C. $2 < x < 5$
 - D. $x > 5$
- 40.** The smallest integer that satisfies $x \geq -1$ is
- A. 0
 - B. -1
 - C. 1
 - D. -2

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Solving simple inequalities

Question 2

- (a) Inequalities with fractions and brackets

Question 3

- (a) Illustrating solutions on the number line

Question 4

- (a) Reading and explaining graphs

Question 5

- (a) Real-life problems

Question 6

- (a) Create and solve

Chapter 2: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Angles at a point add up to
 - A. 90°
 - B. 180°
 - C. 360°
 - D. 270°
2. Angles on a straight line add up to
 - A. 180°
 - B. 360°
 - C. 270°
 - D. 90°
3. Vertically opposite angles are
 - A. right angles
 - B. supplementary
 - C. equal
 - D. complementary
4. Find the value of x.
 - A. 70°
 - B. 110°
 - C. 20°
 - D. 290°
5. Find the value of x.
 - A. 65°
 - B. 25°
 - C. 130°
 - D. 115°
6. Find the value of x.
 - A. 95°
 - B. 215°
 - C. 120°
 - D. 145°

- 7.** The 'F-shape' angles formed by a transversal cutting parallel lines are called
- A. alternate angles
 - B. co-interior angles
 - C. corresponding angles
 - D. vertically opposite angles
- 8.** The 'Z-shape' angles formed by a transversal cutting parallel lines are called
- A. co-interior angles
 - B. vertically opposite angles
 - C. corresponding angles
 - D. alternate angles
- 9.** Co-interior ('C-shape') angles between parallel lines are
- A. reflex
 - B. supplementary
 - C. equal
 - D. complementary
- 10.** Lines a and b are parallel. Find x.
- A. 70°
 - B. 100°
 - C. 110°
 - D. 120°
- 11.** Lines a and b are parallel. Find y.
- A. 70°
 - B. 20°
 - C. 55°
 - D. 110°
- 12.** The complement of 35° is
- A. 35°
 - B. 145°
 - C. 55°
 - D. 65°
- 13.** A polygon with five sides is called a
- A. hexagon
 - B. quadrilateral
 - C. heptagon
 - D. pentagon

- 14.** A polygon with six sides is called a
- A. hexagon
 - B. octagon
 - C. nonagon
 - D. pentagon
- 15.** The sum of the interior angles of a triangle is
- A. 360°
 - B. 90°
 - C. 270°
 - D. 180°
- 16.** Using $(n - 2) \times 180^\circ$, the sum of interior angles of a quadrilateral is
- A. 180°
 - B. 360°
 - C. 540°
 - D. 720°
- 17.** The sum of interior angles of a pentagon is
- A. 720°
 - B. 900°
 - C. 360°
 - D. 540°
- 18.** The sum of interior angles of a hexagon is
- A. 720°
 - B. $1,080^\circ$
 - C. 540°
 - D. 900°
- 19.** The sum of interior angles of an octagon is
- A. 720°
 - B. $1,080^\circ$
 - C. $1,260^\circ$
 - D. $1,440^\circ$
- 20.** A polygon whose interior angles sum to 900° has
- A. 7 sides
 - B. 8 sides
 - C. 9 sides
 - D. 6 sides

- 21.** Each interior angle of a regular pentagon is
- A. 144°
 - B. 72°
 - C. 120°
 - D. 108°
- 22.** Each interior angle of a regular hexagon is
- A. 120°
 - B. 108°
 - C. 135°
 - D. 60°
- 23.** Each exterior angle of a regular octagon is
- A. 30°
 - B. 45°
 - C. 90°
 - D. 40°
- 24.** Find x in the quadrilateral.
- A. 100°
 - B. 80°
 - C. 95°
 - D. 90°
- 25.** The exterior angle is 120° . Find x .
- A. 60°
 - B. 75°
 - C. 120°
 - D. 45°
- 26.** Find x in the hexagon.
- A. 120°
 - B. 130°
 - C. 110°
 - D. 100°
- 27.** Congruent triangles have corresponding sides that are
- A. equal
 - B. perpendicular
 - C. proportional
 - D. parallel

- 28.** Similar triangles have corresponding angles that are
- A. proportional
 - B. supplementary
 - C. equal
 - D. complementary
- 29.** The triangles are similar. Find x .
- A. 12
 - B. 8
 - C. 4.5
 - D. 6
- 30.** Two similar triangles have corresponding sides 5 cm and 15 cm. A side of 7 cm in the smaller triangle corresponds to
- A. 21 cm
 - B. 3 cm
 - C. 7 cm
 - D. 14 cm
- 31.** Which condition does NOT guarantee congruent triangles?
- A. SSS
 - B. AAA
 - C. SAS
 - D. ASA
- 32.** The RHS congruence condition applies to
- A. any triangles
 - B. obtuse triangles only
 - C. equilateral triangles only
 - D. right-angled triangles
- 33.** Two similar triangles have sides in the ratio 2 : 3. If the smaller triangle's perimeter is 20 cm, the larger's is
- A. 45 cm
 - B. 13.3 cm
 - C. 30 cm
 - D. 40 cm
- 34.** Two similar triangles have sides in the ratio 2 : 3. The ratio of their areas is
- A. 2 : 3
 - B. 8 : 27
 - C. 6 : 9
 - D. 4 : 9

- 35.** The angle between the hands of a clock at 3 o'clock is
- A. 60°
 - B. 90°
 - C. 180°
 - D. 30°
- 36.** Two angles on a straight line are $2x$ and $3x$. The value of x is
- A. 60
 - B. 30
 - C. 36
 - D. 45
- 37.** Four equal angles at a point each measure
- A. 45°
 - B. 90°
 - C. 60°
 - D. 120°
- 38.** An isosceles triangle has a vertex angle of 40° . Each base angle is
- A. 70°
 - B. 40°
 - C. 55°
 - D. 140°
- 39.** Each exterior angle of a regular polygon is 30° . The polygon has
- A. 30 sides
 - B. 10 sides
 - C. 6 sides
 - D. 12 sides
- 40.** The interior angles of a polygon sum to $1,440^\circ$. The polygon is a
- A. octagon
 - B. nonagon
 - C. decagon
 - D. dodecagon

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Angle properties at a point, on a line and opposite angles

Question 2

(a) Parallel lines and a transversal

Question 3

(a) The polygon angle-sum formula

Question 4

(a) Missing angles in figures

Question 5

(a) Similar and congruent triangles

Question 6

(a) Create and solve

Chapter 3: Ch12

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The point where the three angle bisectors of a triangle meet is called the
 - A. circumcentre
 - B. centroid
 - C. orthocentre
 - D. incentre
2. The incentre of a triangle is equidistant from
 - A. the midpoints of the three sides
 - B. the two longest sides
 - C. the three sides
 - D. the three vertices
3. To locate the incentre of a triangle by construction, the lines drawn are
 - A. the bisectors of any two angles
 - B. the three altitudes
 - C. the perpendicular bisectors of any two sides
 - D. the three medians
4. The radius of the inscribed circle of a triangle equals
 - A. the distance from the incentre to any vertex
 - B. half of the longest side
 - C. the perpendicular distance from the incentre to any side
 - D. a third of the perimeter
5. The point where the perpendicular bisectors of the sides of a triangle meet is called the
 - A. centroid
 - B. circumcentre
 - C. midpoint
 - D. incentre
6. The circumcentre of a triangle is equidistant from
 - A. the incentre
 - B. the three sides
 - C. the three vertices
 - D. the midpoints of the sides

- 7.** The circle that passes through all three vertices of a triangle is its
- A. inscribed circle
 - B. circumscribed circle
 - C. semicircle
 - D. unit circle
- 8.** The perpendicular bisector of segment AB is the locus of points
- A. equidistant from A and B
 - B. on the segment AB
 - C. equidistant from two parallel lines
 - D. 2 cm from A
- 9.** The bisector of $\angle ABC$ is the locus of points
- A. midway between A and C
 - B. equidistant from B and C
 - C. equidistant from the arms BA and BC
 - D. a fixed distance from A
- 10.** The minimum number of angle bisectors needed to locate the incentre of a triangle is
- A. 0
 - B. 1
 - C. 3
 - D. 2
- 11.** The minimum number of perpendicular bisectors needed to locate the circumcentre of a triangle is
- A. 3
 - B. 0
 - C. 2
 - D. 1
- 12.** The instruments used to construct the perpendicular bisector of a line segment are
- A. a set square and a protractor
 - B. a protractor and a ruler
 - C. a ruler only
 - D. a pair of compasses and a ruler
- 13.** In triangle ABC, I is the incentre and the perpendicular distances from I to each side are 3 cm. The radius of the inscribed circle is
- A. 1.5 cm
 - B. 3 cm
 - C. 6 cm
 - D. 9 cm

- 14.** S is the centre of the circle passing through P, Q and R. SP is equal to
- A. half of PQ
 - B. the radius of the inscribed circle
 - C. SQ
 - D. the perpendicular distance from S to PQ
- 15.** The circumcentre of a right-angled triangle lies at
- A. a point outside the triangle
 - B. the midpoint of the hypotenuse
 - C. the right-angle vertex
 - D. the incentre
- 16.** The circumradius of a right-angled triangle whose hypotenuse is 10 cm is
- A. 5 cm
 - B. 20 cm
 - C. 10 cm
 - D. 2.5 cm
- 17.** The circumcentre lies OUTSIDE the triangle when the triangle is
- A. right-angled
 - B. obtuse-angled
 - C. equilateral
 - D. acute-angled
- 18.** In an equilateral triangle, the incentre and the circumcentre
- A. are the same point
 - B. both lie outside the triangle
 - C. lie on different sides
 - D. are the ends of one side
- 19.** A square ABCD is constructed with $AB = 6.5$ cm. Measuring the diagonal AC gives about
- A. 4.6 cm
 - B. 9.2 cm
 - C. 13.0 cm
 - D. 6.5 cm
- 20.** To construct a square from a given diagonal of 8 cm, one should first
- A. bisect the 8 cm line into four equal parts
 - B. construct a 60° angle at one end
 - C. draw a side of 8 cm
 - D. draw the 8 cm diagonal and construct its perpendicular bisector

- 21.** Each side of the square in question 20 is approximately
- A. 16 cm
 - B. 11.3 cm
 - C. 5.7 cm
 - D. 4.0 cm
- 22.** The minimum data needed to construct a rectangle are
- A. one side only
 - B. the lengths of two adjacent sides
 - C. one diagonal only
 - D. one side and one angle
- 23.** A rectangle has length 6 cm and diagonal 10 cm. Its width is
- A. 12 cm
 - B. 16 cm
 - C. 4 cm
 - D. 8 cm
- 24.** The diagonal of a rectangle with sides 8 cm and 15 cm is
- A. 17 cm
 - B. 15 cm
 - C. 23 cm
 - D. 12 cm
- 25.** The property used when constructing a rhombus with a given side is that a rhombus has
- A. equal diagonals
 - B. all angles equal
 - C. only one pair of parallel sides
 - D. all four sides equal
- 26.** A rhombus has diagonals 6 cm and 8 cm. Each side of the rhombus is
- A. 5 cm
 - B. 10 cm
 - C. 14 cm
 - D. 7 cm
- 27.** A parallelogram can be constructed uniquely when given
- A. one angle only
 - B. two adjacent sides and the included angle
 - C. one side only
 - D. one diagonal only

- 28.** Using the SSS condition with sides 5 cm, 6 cm and 7 cm, the number of different (non-congruent) triangles that can be constructed is
- A. 2
 - B. 3
 - C. infinitely many
 - D. 1
- 29.** Which of the following sets of lengths CANNOT form a triangle?
- A. 5 cm, 12 cm, 13 cm
 - B. 3 cm, 4 cm, 8 cm
 - C. 6 cm, 8 cm, 10 cm
 - D. 3 cm, 4 cm, 5 cm
- 30.** A triangle has sides 5 cm and 8 cm, with the third side a whole number x cm. The least possible value of x is
- A. 13
 - B. 2
 - C. 4
 - D. 3
- 31.** Constructing triangle ABC with $AB = 5$ cm, $BC = 6$ cm and $\angle ABC = 60^\circ$ uses the
- A. SSS condition
 - B. SAS condition
 - C. ASA condition
 - D. RHS condition
- 32.** Constructing a triangle with angles 60° and 60° and the included side 4 cm uses the
- A. ASA condition
 - B. SAS condition
 - C. RHS condition
 - D. SSS condition
- 33.** When constructing a regular hexagon inside a circle of radius r , each side of the hexagon equals
- A. πr
 - B. $2r$
 - C. $r/2$
 - D. r
- 34.** A regular hexagon is constructed inside a circle of radius 4 cm. The perimeter of the hexagon is
- A. 24 cm
 - B. 8 cm
 - C. 16 cm

D. 32 cm

- 35.** A square of side 4 cm has an equilateral triangle attached on top. The perimeter of the compound shape is
- A. 24 cm
 - B. 28 cm
 - C. 20 cm
 - D. 16 cm
- 36.** The locus of a point at a fixed distance of 4 cm from a fixed point O is
- A. a circle of radius 4 cm about O
 - B. the perpendicular bisector of two points
 - C. a straight line
 - D. a pair of parallel lines
- 37.** The locus of a point equidistant from two fixed points P and Q is
- A. a circle centred at P
 - B. the line PQ
 - C. the bisector of angle POQ
 - D. the perpendicular bisector of PQ
- 38.** In constructing the circumscribed circle of a triangle, the line drawn to bisect each side is its
- A. median
 - B. altitude
 - C. perpendicular bisector
 - D. angle bisector
- 39.** Arcs of equal radius from A and B meet at X and Y, and XY bisects AB at right angles. For the construction to work, the radius must be
- A. exactly half of AB
 - B. less than half of AB
 - C. equal to AB
 - D. greater than half of AB
- 40.** A circle drawn inside a square of side 6 cm touches all four sides. The diameter of the circle is
- A. 6 cm
 - B. 3 cm
 - C. 12 cm
 - D. 9 cm

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Constructing a triangle and its inscribed circle

Question 2

- (a) Constructing a triangle and its circumscribed circle

Question 3

- (a) Constructing a square

Question 4

- (a) Constructing a rectangle from a side and a diagonal

Question 5

- (a) Parallelogram and rhombus

Question 6

- (a) Equilateral triangle: inscribed and circumscribed circles

Chapter 4: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The cross-section of a prism is
 - A. the same shape and size along its length
 - B. a circle that widens along the length
 - C. a shape that changes along the length
 - D. a single point at each end
2. The number of faces of a cuboid is
 - A. 8
 - B. 6
 - C. 12
 - D. 4
3. The number of vertices of a cuboid is
 - A. 8
 - B. 12
 - C. 6
 - D. 10
4. The number of edges of solid P is
 - A. 6
 - B. 8
 - C. 12
 - D. 9
5. Solid Q is a
 - A. pyramid
 - B. cone
 - C. cuboid
 - D. triangular prism
6. The net of a cuboid is made up of
 - A. 4 rectangles and 2 squares
 - B. 6 squares
 - C. 6 rectangles
 - D. 4 rectangles and 2 triangles

- 7.** The net of a triangular prism is made up of
- A. 3 triangles and 2 rectangles
 - B. 2 triangles and 4 rectangles
 - C. 6 rectangles
 - D. 2 triangles and 3 rectangles
- 8.** A solid with 5 faces, 9 edges and 6 vertices is a
- A. triangular prism
 - B. square-based pyramid
 - C. cone
 - D. cuboid
- 9.** When a cuboid of length l , width w and height h is unfolded, the six rectangles have a total area of
- A. $l \times w \times h$
 - B. $lw + lh + wh$
 - C. $2(lw + lh + wh)$
 - D. $4(lw + lh + wh)$
- 10.** The lateral (side) surface area of any prism equals
- A. area of cross-section $\times$ length
 - B. perimeter of cross-section $\times$ length
 - C. $2 \times$ area of cross-section $\times$ length
 - D. half the perimeter $\times$ length
- 11.** The total length of all the edges of a cuboid l , w , h is
- A. $l + w + h$
 - B. $6(l + w + h)$
 - C. $4(l + w + h)$
 - D. $2(l + w + h)$
- 12.** A prism whose base is an n -sided polygon has how many rectangular lateral faces?
- A. $n - 2$
 - B. n
 - C. $n + 2$
 - D. $2n$
- 13.** The surface area of a cuboid $4 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}$ is
- A. 52 cm^2
 - B. 24 cm^2
 - C. 40 cm^2
 - D. 94 cm^2

- 14.** The surface area of a cube of side 4 cm is
- A. 48 cm^2
 - B. 64 cm^2
 - C. 16 cm^2
 - D. 96 cm^2
- 15.** The surface area of a cube is 150 cm^2 . Its side is
- A. 10 cm
 - B. 25 cm
 - C. 5 cm
 - D. 6 cm
- 16.** The surface area of a cuboid $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$ is
- A. 60 cm^2
 - B. 84 cm^2
 - C. 120 cm^2
 - D. 94 cm^2
- 17.** The net folds into a cuboid. Its surface area is
- A. 108 cm^2
 - B. 72 cm^2
 - C. 96 cm^2
 - D. 54 cm^2
- 18.** An open box with no lid has base $6 \text{ cm} \times 4 \text{ cm}$ and height 5 cm. The total area of its five faces is
- A. 120 cm^2
 - B. 148 cm^2
 - C. 100 cm^2
 - D. 124 cm^2
- 19.** A cuboid has length 6 cm, width 4 cm and surface area 148 cm^2 . Its height is
- A. 4 cm
 - B. 5 cm
 - C. 6 cm
 - D. 7 cm
- 20.** The surface area of a cuboid $10 \text{ cm} \times 8 \text{ cm} \times 5 \text{ cm}$ is
- A. 340 cm^2
 - B. 230 cm^2
 - C. 400 cm^2
 - D. 170 cm^2

- 21.** When the side of a cube is doubled, its surface area becomes
- A. the same
 - B. 2 times as large
 - C. 4 times as large
 - D. 8 times as large
- 22.** A gift box measures 20 cm $\times$ 15 cm $\times$ 10 cm. Ignoring overlaps, the wrapping paper needed to cover it completely is
- A. 700 cm²
 - B. 1 300 cm²
 - C. 650 cm²
 - D. 3 000 cm²
- 23.** The four walls of a room 5 m $\times$ 4 m and height 3 m are to be painted. The painted area is
- A. 60 m²
 - B. 94 m²
 - C. 54 m²
 - D. 47 m²
- 24.** Two cubes of side 2 cm are joined face to face to form a cuboid. The surface area of the cuboid is
- A. 32 cm²
 - B. 40 cm²
 - C. 48 cm²
 - D. 44 cm²
- 25.** A prism has triangular cross-section of area A and perimeter P, with length l. Its surface area is
- A. $A + Pl$
 - B. $2A + \frac{1}{2}Pl$
 - C. $2A + Pl$
 - D. $2(A + Pl)$
- 26.** A triangular prism has right-triangle cross-section with sides 3 cm, 4 cm and 5 cm, and length 10 cm. Its surface area is
- A. 120 cm²
 - B. 144 cm²
 - C. 96 cm²
 - D. 132 cm²
- 27.** The net folds into a triangular prism. Its surface area is
- A. 120 cm²
 - B. 108 cm²
 - C. 84 cm²
 - D. 96 cm²

- 28.** A prism has isosceles-triangle cross-section with base 8 cm, height 3 cm and equal sides 5 cm. Its length is 12 cm. Its surface area is
- A. 240 cm^2
 - B. 216 cm^2
 - C. 228 cm^2
 - D. 264 cm^2
- 29.** A triangular prism has right-triangle cross-section 6 cm–8 cm–10 cm and length 20 cm. Its surface area is
- A. 480 cm^2
 - B. 528 cm^2
 - C. 576 cm^2
 - D. 432 cm^2
- 30.** The tent is a triangular prism with isosceles triangular ends (base 2 m, height 0.75 m, slant sides 1.25 m) and length 3 m. The cloth for the two ends and the two slanted faces (no floor) is
- A. 9 m^2
 - B. 10.5 m^2
 - C. 7.5 m^2
 - D. 8.25 m^2
- 31.** A net of 2 triangles and 3 rectangles folds into a
- A. cuboid
 - B. pyramid
 - C. cone
 - D. triangular prism
- 32.** A triangular prism has equilateral-triangle cross-section of side 6 cm (triangle height ≈ 5.2 cm) and length 10 cm. Its surface area is about
- A. 195.6 cm^2
 - B. 211.2 cm^2
 - C. 226.8 cm^2
 - D. 180 cm^2
- 33.** A prism has triangular cross-section of area 24 cm^2 and perimeter 24 cm. Its length is 10 cm. Its surface area is
- A. 240 cm^2
 - B. 264 cm^2
 - C. 288 cm^2
 - D. 312 cm^2

- 34.** In the net of a triangular prism, the two congruent triangles become the
- A. lateral faces of the prism
 - B. diagonals of the prism
 - C. vertices of the prism
 - D. bases (ends) of the prism
- 35.** The number of faces of a triangular prism is
- A. 6
 - B. 7
 - C. 5
 - D. 4
- 36.** Surface area of a solid is measured in
- A. units
 - B. cubic units
 - C. units of length
 - D. square units
- 37.** The difference between the surface areas of a cube of side 4 cm and a cuboid $6\text{ cm} \times 4\text{ cm} \times 2\text{ cm}$ is
- A. 8 cm^2
 - B. 4 cm^2
 - C. 16 cm^2
 - D. 24 cm^2
- 38.** The net of a prism with a pentagonal base contains how many rectangles?
- A. 6
 - B. 5
 - C. 7
 - D. 3
- 39.** A cuboid has a square base of side 3 cm and height 8 cm. Its surface area is
- A. 132 cm^2
 - B. 96 cm^2
 - C. 108 cm^2
 - D. 114 cm^2
- 40.** The total edge length of a cube is 60 cm. Its surface area is
- A. 150 cm^2
 - B. 100 cm^2
 - C. 125 cm^2
 - D. 300 cm^2

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Identifying solids and drawing nets

Question 2

(a) Deriving the cuboid surface-area formula

Question 3

(a) Cuboid surface-area problems

Question 4

(a) Deriving the triangular-prism surface area

Question 5

(a) Triangular-prism problems

Question 6

(a) Applications

Chapter 5: Ch14

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. With O as the origin, the position vector of M is
 - A. $(3, 2)$
 - B. $(-2, -3)$
 - C. $(2, -3)$
 - D. $(2, 3)$
2. With O as the origin, the position vector of N is
 - A. $(2, -1)$
 - B. $(-1, 2)$
 - C. $(-2, 1)$
 - D. $(1, -2)$
3. A movement of 3 units right and 2 units down is written as the column vector
 - A. $(3, 2)$
 - B. $(-3, 2)$
 - C. $(2, 3)$
 - D. $(3, -2)$
4. The magnitude of the vector $(3, 4)$ is
 - A. 7
 - B. 5
 - C. 12
 - D. 25
5. The vector that translates A(1, 2) to B(4, 6) is
 - A. $(3, 4)$
 - B. $(-3, -4)$
 - C. $(4, 3)$
 - D. $(5, 8)$
6. The negative of the vector $(-3, 4)$ is
 - A. $(3, 4)$
 - B. $(-3, -4)$
 - C. $(3, -4)$
 - D. $(4, -3)$

- 7.** Two vectors are equal when they have
- A. the same magnitude and the same direction
 - B. the same magnitude only
 - C. the same direction only
 - D. the same starting point
- 8.** $2 \times (2, -3) =$
- A. $(2, -6)$
 - B. $(4, 6)$
 - C. $(4, -3)$
 - D. $(4, -6)$
- 9.** Two vectors are parallel when
- A. their dot product is zero
 - B. their magnitudes are equal
 - C. one is a scalar multiple of the other
 - D. they start from the same point
- 10.** The vector $(6, 8)$ is parallel to
- A. $(4, 3)$
 - B. $(3, 4)$
 - C. $(-3, 4)$
 - D. $(8, 6)$
- 11.** If $(k, 6)$ is parallel to $(2, 3)$, then $k =$
- A. 6
 - B. 9
 - C. 4
 - D. 2
- 12.** The vectors $(3x, 2)$ and $(6, x)$ are parallel when x is
- A. 12
 - B. 2 or -2
 - C. 4
 - D. 6
- 13.** Two vectors are perpendicular when
- A. their dot product is zero
 - B. their magnitudes are equal
 - C. one is a scalar multiple of the other
 - D. they have the same direction

- 14.** A vector perpendicular to $(3, 4)$ is
- A. $(4, 3)$
 - B. $(3, -4)$
 - C. $(-4, 3)$
 - D. $(-3, 4)$
- 15.** A vector perpendicular to $(2, 5)$ is
- A. $(5, -2)$
 - B. $(5, 2)$
 - C. $(-2, -5)$
 - D. $(2, -5)$
- 16.** The dot product of (a, b) and (c, d) is
- A. $ad + bc$
 - B. $ab + cd$
 - C. $ac - bd$
 - D. $ac + bd$
- 17.** If $(x, 4)$ is perpendicular to $(2, -1)$, then $x =$
- A. 8
 - B. 4
 - C. 2
 - D. -2
- 18.** If $(p, 3)$ is parallel to $(4, 12)$, then $p =$
- A. 9
 - B. 3
 - C. 4
 - D. 1
- 19.** The vectors $(2, -3)$ and $(-4, 6)$ are
- A. perpendicular
 - B. parallel
 - C. equal
 - D. neither parallel nor perpendicular
- 20.** The vectors $(1, 2)$ and $(2, 1)$ are
- A. perpendicular
 - B. equal
 - C. neither parallel nor perpendicular
 - D. parallel

- 21.** If $(x, 2)$ is parallel to $(3, 6)$, then $x =$
- A. 1
 - B. 6
 - C. 2
 - D. 3
- 22.** A vector perpendicular to $(-3, 4)$ is
- A. $(3, 4)$
 - B. $(-4, -3)$
 - C. $(4, 3)$
 - D. $(4, -3)$
- 23.** The triangle law of vector addition states that $AB + BC =$
- A. AC
 - B. CA
 - C. BA
 - D. CB
- 24.** Using the grid, $a + b =$
- A. $(1, -5)$
 - B. $(5, -1)$
 - C. $(5, 1)$
 - D. $(-5, 1)$
- 25.** $(2, 5) - (1, 2) =$
- A. $(1, 3)$
 - B. $(3, 7)$
 - C. $(1, -3)$
 - D. $(3, 3)$
- 26.** If $P(1, -3)$ and $Q(7, 5)$, the column vector PQ is
- A. $(-6, -8)$
 - B. $(6, 8)$
 - C. $(6, -8)$
 - D. $(8, 6)$
- 27.** The magnitude of PQ in question 26 is
- A. 100
 - B. 14
 - C. 8
 - D. 10

- 28.** Triangle PQR with $P(1, -3)$, $Q(7, 5)$, $R(-3, 5)$ is isosceles because
- A. $PQ = QR = 10$ units
 - B. $PQ = PR$
 - C. $QR = PR$
 - D. all sides are equal
- 29.** In the parallelogram law, the resultant of u and v is the vector along
- A. the shorter side
 - B. the diagonal from the common start point
 - C. either dashed side
 - D. the longer of u and v
- 30.** The resultant of $(2, 1)$ and $(3, 4)$ is
- A. $(5, 5)$
 - B. $(5, 4)$
 - C. $(1, 3)$
 - D. $(6, 4)$
- 31.** If $a = (2, 3)$ and $b = (1, -1)$, then $2a + b =$
- A. $(3, 5)$
 - B. $(5, 2)$
 - C. $(4, 4)$
 - D. $(5, 5)$
- 32.** If $a = (1, 2)$ and $b = (2, 1)$, then $3a - 2b =$
- A. $(-1, -4)$
 - B. $(7, 4)$
 - C. $(-1, 4)$
 - D. $(1, 4)$
- 33.** In parallelogram PQRS, the vector PQ is equal to
- A. SR
 - B. QS
 - C. RS
 - D. QR
- 34.** PQRS is a parallelogram with $P(x, y)$, $Q(5, 7)$, $R(2, 4)$ and $S(1, 3)$. The values of x and y are
- A. $x = 6, y = 4$
 - B. $x = 3, y = 5$
 - C. $x = 4, y = 6$
 - D. $x = 5, y = 6$

- 35.** A boat sails $(3, 4)$ and then $(-1, 2)$. Its total displacement is
- A. $(2, 2)$
 - B. $(-4, -6)$
 - C. $(4, 2)$
 - D. $(2, 6)$
- 36.** The magnitude of the resultant of $(3, 0)$ and $(0, 4)$ is
- A. 25
 - B. 5
 - C. 7
 - D. 12
- 37.** If $\vec{AB} = (2, 5)$ and A is $(1, 1)$, then B is
- A. $(1, -4)$
 - B. $(-1, -4)$
 - C. $(6, 3)$
 - D. $(3, 6)$
- 38.** If M is the midpoint of PQ, then PM equals
- A. PQ
 - B. $-\frac{1}{2}$ PQ
 - C. $\frac{1}{2}$ PQ
 - D. 2 PQ
- 39.** $(2, 3) + (-2, -3)$ equals
- A. $(-4, -6)$
 - B. $(4, 6)$
 - C. $(0, 6)$
 - D. the zero vector $(0, 0)$
- 40.** A displacement of 5 km due east of O, written as a column vector, is
- A. $(0, 5)$
 - B. $(5, 0)$
 - C. $(-5, 0)$
 - D. $(5, 5)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Position vectors and notation

Question 2

- (a) Parallel vectors

Question 3

- (a) Perpendicular vectors

Question 4

- (a) The triangle law of addition

Question 5

- (a) The parallelogram law of addition

Question 6

- (a) Vector journeys

Chapter 6: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. An enlargement is completely defined by
 - A. a centre and a scale factor
 - B. an angle and a distance
 - C. a line of symmetry
 - D. a direction and a length
2. Under an enlargement, the angles of the object are
 - A. halved
 - B. multiplied by the scale factor
 - C. unchanged
 - D. doubled
3. Under an enlargement of scale factor k , the lengths of the object are
 - A. unchanged
 - B. multiplied by k^2
 - C. increased by k
 - D. multiplied by $|k|$
4. Under an enlargement of scale factor k , the area of the object is multiplied by
 - A. k^2
 - B. k
 - C. $2k$
 - D. $\sqrt{k}$
5. An enlargement with scale factor 1 gives an image that is
 - A. smaller than the object
 - B. a mirror image of the object
 - C. twice the object
 - D. congruent to the object
6. An enlargement with scale factor -1 gives an image that is
 - A. a smaller copy of the object
 - B. a larger copy of the object
 - C. the object turned through 180° about the centre
 - D. a translation of the object

- 7.** The lines joining corresponding vertices of an object and its image under an enlargement all pass through
- A. the origin
 - B. the midpoint of the object
 - C. the image
 - D. the centre of enlargement
- 8.** An enlargement with scale factor 2 changes the perimeter of the object by a factor of
- A. 8
 - B. 2
 - C. $\sqrt{2}$
 - D. 4
- 9.** Which situation is an example of enlargement?
- A. flipping a page of a book
 - B. turning a door on its hinges
 - C. printing a small photograph as a poster
 - D. sliding a box across the floor
- 10.** A projector throws an image on a screen. The image is an enlargement because it is
- A. the same size but a different shape
 - B. the same shape but a different size
 - C. a reflection of the slide
 - D. a translation of the slide
- 11.** The shadow of a triangle on a wall has the same shape but a larger size. This is an example of
- A. enlargement
 - B. rotation
 - C. translation
 - D. reflection
- 12.** Which of the following is NOT an enlargement?
- A. blowing up a passport photo to poster size
 - B. a reflection in a plane mirror
 - C. zooming in on a phone screen
 - D. the shadow of a chair at sunset
- 13.** Photocopying a document at 150% corresponds to a scale factor of
- A. 1.5
 - B. 0.15
 - C. 15
 - D. 150

- 14.** Zooming $\times 3$ on a phone map corresponds to a scale factor of
- A. 0.3
 - B. 30
 - C. $\frac{1}{3}$
 - D. 3
- 15.** An object length of 4 cm has an image length of 12 cm. The scale factor is
- A. 3
 - B. $\frac{1}{3}$
 - C. 8
 - D. 48
- 16.** An object length of 6 cm has an image length of 3 cm. The scale factor is
- A. 2
 - B. $\frac{1}{2}$
 - C. 3
 - D. 18
- 17.** The image lies on the opposite side of the centre and is twice as large as the object. The scale factor is
- A. $-\frac{1}{2}$
 - B. $\frac{1}{2}$
 - C. -2
 - D. 2
- 18.** The scale factor of the enlargement that maps P onto Q is
- A. 4
 - B. 2
 - C. $\frac{1}{2}$
 - D. -2
- 19.** The scale factor of the enlargement that maps P onto Q is
- A. 2
 - B. $-\frac{1}{2}$
 - C. $\frac{1}{2}$
 - D. -2
- 20.** The scale factor of the enlargement that maps P onto Q is
- A. -2
 - B. $\frac{1}{2}$
 - C. $-\frac{1}{2}$
 - D. 2

- 21.** The image of $(2, 3)$ under an enlargement with centre O and scale factor 3 is
- A. $(9, 6)$
 - B. $(5, 6)$
 - C. $(2, 9)$
 - D. $(6, 9)$
- 22.** The image of $(4, -2)$ under an enlargement with centre O and scale factor -2 is
- A. $(-8, 4)$
 - B. $(8, -4)$
 - C. $(-8, -4)$
 - D. $(2, -4)$
- 23.** An enlargement with centre $(1, 1)$ and scale factor 2 maps $(3, 1)$ to
- A. $(7, 1)$
 - B. $(4, 2)$
 - C. $(5, 1)$
 - D. $(6, 2)$
- 24.** When $0 < k < 1$, the image is
- A. larger and on the opposite side of the centre
 - B. larger and on the same side of the centre
 - C. smaller and on the opposite side of the centre
 - D. smaller and on the same side of the centre
- 25.** When $k < -1$, the image is
- A. larger and on the same side of the centre
 - B. smaller and on the opposite side of the centre
 - C. larger and on the opposite side of the centre
 - D. smaller and on the same side of the centre
- 26.** Under an enlargement with centre O and scale factor k , the point (x, y) maps to
- A. (kx, y)
 - B. (kx, ky)
 - C. $(x/k, y/k)$
 - D. $(x + k, y + k)$
- 27.** The ratio of the perimeter of the image to the perimeter of the object equals
- A. $1/|k|$
 - B. k^2
 - C. $2k$
 - D. $|k|$

- 28.** An object has area 10 cm^2 . With scale factor 3, the image area is
- A. 90 cm^2
 - B. 30 cm^2
 - C. 13 cm^2
 - D. 900 cm^2
- 29.** An image has area 4 cm^2 and the object has area 36 cm^2 . A possible scale factor is
- A. 9
 - B. $\frac{1}{3}$ or $-\frac{1}{3}$
 - C. 3 or -3
 - D. $\frac{1}{9}$
- 30.** The centre of the enlargement that maps P onto Q is
- A. (1, 0)
 - B. (1, 1)
 - C. (0, 0)
 - D. (2, 2)
- 31.** Congruent shapes have
- A. the same shape but different sizes
 - B. the same shape and the same size
 - C. different shapes but the same size
 - D. different shapes and sizes
- 32.** Similar shapes have
- A. equal areas
 - B. equal perimeters
 - C. equal angles and proportional sides
 - D. equal sides only
- 33.** The image under any enlargement is always
- A. similar to the object
 - B. a mirror image of the object
 - C. congruent to the object
 - D. larger than the object
- 34.** The image of an enlargement is congruent to the object when
- A. $k = 2$
 - B. $k = 0$
 - C. $|k| = 1$
 - D. $k > 1$

- 35.** Similar triangles with sides in the ratio 2 : 3 have areas in the ratio
- A. 8 : 27
 - B. 4 : 9
 - C. $\sqrt{2} : \sqrt{3}$
 - D. 2 : 3
- 36.** Two similar rectangles have shorter sides 3 cm and 9 cm. If the longer side of the smaller is 5 cm, the longer side of the larger is
- A. 15 cm
 - B. 45 cm
 - C. 11 cm
 - D. 25 cm
- 37.** Which pair of shapes is ALWAYS similar?
- A. any two rectangles
 - B. any two isosceles triangles
 - C. any two rhombuses
 - D. any two squares
- 38.** Which pair of shapes is NOT necessarily similar?
- A. any two rectangles
 - B. any two equilateral triangles
 - C. any two squares
 - D. any two circles
- 39.** Similar shapes with scale factor 4 have areas in the ratio
- A. 1 : 64
 - B. 1 : 4
 - C. 1 : 16
 - D. 1 : 8
- 40.** An object with sides 2 cm and 3 cm maps to a shape with sides 6 cm and 9 cm. The shapes are
- A. congruent
 - B. similar with scale factor 9
 - C. neither similar nor congruent
 - D. similar with scale factor 3

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Everyday enlargements and properties

Question 2

(a) Drawing enlargements on a grid

Question 3

(a) Scale factors from diagrams

Question 4

(a) Enlargement coordinates and areas

Question 5

(a) Congruent and similar shapes

Question 6

(a) Verifying similarity

Chapter 7: Ch16

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Quantitative data are
 - A. descriptive data expressed in words
 - B. data taken from existing records
 - C. numerical data obtained by counting or measuring
 - D. data collected by interview only
2. Qualitative data are
 - A. descriptive (non-numerical) data
 - B. grouped data
 - C. secondary data
 - D. numerical data
3. Musa wants to find the books that are most popular among his friends. The best method of collecting the data is
 - A. using a database
 - B. an experiment
 - C. a questionnaire or survey
 - D. observation
4. To find the most common mode of travel by learners in a school, the best method is
 - A. an experiment
 - B. observation only
 - C. measurement with instruments
 - D. a survey using a questionnaire
5. Counting vehicles passing a junction between 7 am and 8 am is best done by
 - A. observation
 - B. questionnaire
 - C. experiment
 - D. interview
6. Life-testing thirty bulbs to the nearest hour is an example of data collection by
 - A. using existing records
 - B. survey
 - C. interview

D. experiment

- 7.** Asking respondents questions face to face and recording their answers is called
- A. an interview
 - B. a questionnaire
 - C. observation
 - D. an experiment
- 8.** Using the school's attendance register, kept over the years, to study attendance is using
- A. observation
 - B. existing records (secondary data)
 - C. an interview
 - D. primary data from an experiment
- 9.** The correct tally for 13 is
- A. thirteen separate single strokes as standard practice
 - B. one group of five and three strokes
 - C. three complete groups of five
 - D. two complete tally groups of five and three extra strokes
- 10.** The sum of the frequencies in a frequency table always equals
- A. the number of data values
 - B. the range
 - C. the mode
 - D. the number of classes
- 11.** The mode of the data 2, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 7, 7 is
- A. 6
 - B. 7
 - C. 4
 - D. 5
- 12.** The mode of 4, 7, 7, 9, 7, 4 is
- A. 6
 - B. 7
 - C. 4
 - D. 9
- 13.** The range of 12, 5, 9, 3, 8 is
- A. 7
 - B. 12
 - C. 9
 - D. 8

- 14.** The median of 3, 5, 7, 9 is
- A. 7
 - B. 8
 - C. 5
 - D. 6
- 15.** The value represented by $2 \mid 4$ is
- A. 240
 - B. 24
 - C. 42
 - D. 6
- 16.** The number of data values shown is
- A. 9
 - B. 7
 - C. 8
 - D. 10
- 17.** In a bar chart, the bars
- A. are always drawn horizontal
 - B. must show continuous data
 - C. touch each other
 - D. are separated by gaps
- 18.** The representation best suited to showing change over time is a
- A. line graph
 - B. waffle diagram
 - C. pie chart
 - D. pictograph
- 19.** In a pie chart, a frequency of 3 out of a total of 30 gives a sector angle of
- A. 30°
 - B. 36°
 - C. 60°
 - D. 108°
- 20.** In a pictograph one symbol represents 10 books. Four and a half symbols represent
- A. 50 books
 - B. 14 books
 - C. 45 books
 - D. 40 books

- 21.** In a 10×10 waffle diagram, each small square represents
- A. 5%
 - B. 1%
 - C. 10%
 - D. 0.1%
- 22.** A box-and-whisker plot displays the median together with
- A. the mode only
 - B. the frequency of every value
 - C. the mean only
 - D. the quartiles and extreme values
- 23.** The class width of the class interval 168–171 (whole-number data) is
- A. 5
 - B. 4
 - C. 8
 - D. 3
- 24.** A grouped table has frequencies 5, 9, 12 and 7. The modal class is the class with frequency
- A. 7
 - B. 5
 - C. 12
 - D. 9
- 25.** In the data 2, 3, 4, 4, 4, 4, 5, 5, 5, 5, 5, 6, 6, 7, 7 the frequency of 4 is
- A. 3
 - B. 5
 - C. 6
 - D. 4
- 26.** The diagram used to represent continuous grouped data is a
- A. pie chart
 - B. waffle diagram
 - C. histogram
 - D. pictograph
- 27.** In a histogram the bars touch each other because
- A. the data are continuous
 - B. the bars must be equal in height
 - C. the data are categorical
 - D. all frequencies are equal

- 28.** A frequency table has frequencies 3, x , 6, 5, 2 with a total of 20. The value of x is
- A. 3
 - B. 6
 - C. 4
 - D. 8
- 29.** In a histogram, the mode is read from
- A. the last bar
 - B. the middle bar
 - C. the shortest bar
 - D. the tallest bar
- 30.** The modal class is
- A. 30-39
 - B. 50-59
 - C. 10-19
 - D. 20-29
- 31.** The total number of students is
- A. 25
 - B. 30
 - C. 35
 - D. 10
- 32.** The frequency of the class 20-29 is
- A. 6
 - B. 10
 - C. 4
 - D. 7
- 33.** The difference between the greatest and the least frequency is
- A. 10
 - B. 7
 - C. 3
 - D. 4
- 34.** A histogram has no gaps between its bars because the data are
- A. categorical
 - B. discrete categories
 - C. continuous (grouped measurements)
 - D. qualitative

- 35.** For grouped data, the mode is given as
- A. the modal class
 - B. the midpoint of the first class
 - C. half of the greatest frequency
 - D. a single exact value
- 36.** Bulb lifespans (hours): 164–167 $f = 4$; 168–171 $f = 7$; 172–175 $f = 11$; 176–179 $f = 8$. The modal class is
- A. 176–179
 - B. 164–167
 - C. 172–175
 - D. 168–171
- 37.** If the mode of the data 2, 4, 4, x , 6 is 4, then x could be
- A. 2
 - B. 4
 - C. 6
 - D. any number
- 38.** The class that contains the median mark is
- A. 30–39
 - B. 20–29
 - C. 40–49
 - D. 10–19
- 39.** Five classes have frequencies 2, 5, 8, 5, 2. The modal class is
- A. the fifth class
 - B. the third class
 - C. the first class
 - D. the second class
- 40.** A histogram has total frequency 40. Four bars show 8, 12, 10 and 6. The fifth bar shows
- A. 6
 - B. 10
 - C. 4
 - D. 14

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Data types and collection methods

Question 2

- (a) Ungrouped frequency table

Question 3

- (a) Grouped frequency table

Question 4

- (a) Stem-and-leaf, pictograph and pie chart

Question 5

- (a) Histograms

Question 6

- (a) Create and solve

Chapter 8: Ch1

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. A survey question 'Overall, don't you think the teaching of mathematics is very good?' is biased because it
 - A. is too short
 - B. uses difficult words
 - C. leads respondents towards answering 'yes'
 - D. has too many options
2. A better, neutral version of the question is 'Overall, how will you rate the teaching of mathematics?' with the scale
 - A. agree - disagree only
 - B. very poor - poor - fair - good - very good
 - C. true - false
 - D. yes - no
3. Ama surveyed only the school's football team about sport-related injuries. The main problem is that
 - A. the sample is only footballers (selection bias)
 - B. injuries never happen
 - C. football is not a sport
 - D. the team is too small to exist
4. To remove the bias, Ama should ask
 - A. more footballers only
 - B. only the coaches
 - C. people in a variety of sports, contact and non-contact
 - D. only spectators
5. According to the curriculum, which group of factors may introduce bias into data collection?
 - A. mean, median and mode
 - B. language, ethics, cost, time and timing, privacy, cultural sensitivity
 - C. bar charts and pie charts
 - D. rulers, compasses and protractors

- 6.** Asking 'If you eat pork, name your favourite method of cooking it' in a community where pork is forbidden may offend respondents. The influencing factor is
- A. privacy
 - B. cost
 - C. cultural sensitivity
 - D. timing
- 7.** A survey on market sales conducted only between 7 am and 9 am is likely to be biased because of
- A. timing — morning traders are not representative of the whole day
 - B. the cost of printing
 - C. the language used
 - D. the privacy of traders
- 8.** Asking respondents to state their exact personal income in a public place raises the issue of
- A. timing
 - B. sample size
 - C. cost
 - D. privacy
- 9.** The mean of 4, 6, 8, 10, 12 is
- A. 40
 - B. 8
 - C. 6
 - D. 10
- 10.** The mean of 5, 7, 9 is
- A. 6
 - B. 8
 - C. 21
 - D. 7
- 11.** The median of 3, 8, 5, 2, 9 is
- A. 8
 - B. 27
 - C. 5
 - D. 3
- 12.** The mode of 2, 3, 3, 4, 5, 3, 2 is
- A. 5
 - B. 3
 - C. 2
 - D. 4

- 13.** The range of 14, 3, 9, 20, 7 is
- A. 11
 - B. 20
 - C. 53
 - D. 17
- 14.** Bulb lifespans were recorded to the nearest hour from 164 to 179. The minimum and maximum lifespans are
- A. 164 hours and 179 hours
 - B. 164 hours and 176 hours
 - C. 168 hours and 179 hours
 - D. 167 hours and 176 hours
- 15.** The range of the bulb lifespans 164–179 hours is
- A. 343 hours
 - B. 14 hours
 - C. 15 hours
 - D. 16 hours
- 16.** Values 2, 3, 4 occur with frequencies 3, 5, 2. The mean is
- A. 3.0
 - B. 2.5
 - C. 9.0
 - D. 2.9
- 17.** The sum of n values equals
- A. $\text{mean} \div n$
 - B. $\text{mean} \times n$
 - C. $n \div \text{mean}$
 - D. $\text{mean} + n$
- 18.** The mean of four numbers is 6. If three of them are 5, 7 and 4, the fourth is
- A. 24
 - B. 6
 - C. 10
 - D. 8
- 19.** The median of 4, 6, 8, 10 is
- A. 7
 - B. 6
 - C. 8
 - D. 28

- 20.** The mode of the data is
- A. 3 siblings
 - B. 20 siblings
 - C. 2 siblings
 - D. 8 siblings
- 21.** When a data set contains extreme values, the most reliable measure of central tendency is usually the
- A. median
 - B. sum
 - C. mean
 - D. range
- 22.** The measure of central tendency that uses every value in the data set is the
- A. median
 - B. mode
 - C. range
 - D. mean
- 23.** The mean of the first five natural numbers 1, 2, 3, 4, 5 is
- A. 3
 - B. 3.5
 - C. 15
 - D. 2.5
- 24.** The mean age of 10 people is 15 years. A 26-year-old joins them. The new mean is
- A. 41 years
 - B. 16 years
 - C. 15 years
 - D. 17 years
- 25.** A shopkeeper ordering bulbs to sell would mostly rely on which statistic, to stock the lifespan customers most often get?
- A. the maximum
 - B. the minimum
 - C. the mode
 - D. the range
- 26.** The median lifespan of the bulbs tells the shopkeeper that
- A. half of the bulbs lasted at least that long
 - B. most bulbs failed before that time
 - C. the lifespans are all equal
 - D. all bulbs lasted that long

- 27.** The number of students represented is
- A. 40
 - B. 20
 - C. 8
 - D. 24
- 28.** The mean number of siblings is
- A. 2
 - B. 1
 - C. 3
 - D. 4
- 29.** The mean of 2, 3, 3, 5, 7 is 4. If the 7 is replaced by 300, the new mean is
- A. 20.0
 - B. 62.6
 - C. 313
 - D. 4.0
- 30.** The measure of central tendency most affected by an extreme value is the
- A. frequency
 - B. median
 - C. mode
 - D. mean
- 31.** The measure LEAST affected by adding one extreme value to 2, 3, 3, 5, 7 is the
- A. mean
 - B. sum
 - C. median (and the mode)
 - D. range
- 32.** The median of 3, 4, 5, 6, 7 is 5. If 7 is replaced by 100, the new median is
- A. 25
 - B. 6
 - C. 100
 - D. 5
- 33.** Adding one extreme value that occurs only once to a data set leaves which measure unchanged?
- A. the range
 - B. the mean
 - C. the mode
 - D. the sum

- 34.** Data 10, 12, 12, 14, 16: replacing 16 with 160 changes which measure the most?
- A. the mean
 - B. the mode
 - C. the median
 - D. none of them
- 35.** The mean of five values is 8. A sixth value of 20 is added. The new mean is
- A. 8
 - B. 28
 - C. 10
 - D. 14
- 36.** Bulbs with mean lifespan 170 hours: one bulb is replaced by a 300-hour bulb. The mean will
- A. become 300
 - B. decrease
 - C. stay exactly 170
 - D. increase
- 37.** A data set has mean 9, median 9 and mode 9. One extreme value 100 is added. Which measure stays 9?
- A. the mean
 - B. the mode
 - C. the sum
 - D. all of them
- 38.** A value that is far away from the rest of the data is called
- A. an outlier (extreme value)
 - B. a class interval
 - C. a frequency
 - D. a mode
- 39.** Five workers earn GH¢ 900–1 050 but the manager earns GH¢ 10 000. The typical salary is best described by the
- A. sum
 - B. mean
 - C. maximum
 - D. median
- 40.** Data 5, 6, 7, 8, 9 has mean 7. Adding 70 gives
- A. mean 7.5 and median 17.5
 - B. mean 17.5 and median 7.5
 - C. mean 12 and median 7
 - D. mean 7 and median 70

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Bias in data collection

Question 2

- (a) Descriptive statistics of ungrouped data

Question 3

- (a) Descriptive statistics of grouped data

Question 4

- (a) Working with the mean

Question 5

- (a) Effect of extreme data

Question 6

- (a) Interpreting with extreme values

Chapter 9: Ch18

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The sample space for tossing a coin once is
 - A. {Heads, Tails}
 - B. {1, 2}
 - C. {Heads}
 - D. {0, 1, 2}
2. The sample space for rolling a fair die has
 - A. 4 outcomes
 - B. 36 outcomes
 - C. 6 equally likely outcomes
 - D. 2 outcomes
3. The number of outcomes in the sample space when two coins are tossed is
 - A. 16
 - B. 4
 - C. 2
 - D. 8
4. The probability of any event lies between
 - A. 0 and 1
 - B. -1 and 1
 - C. 0 and 100
 - D. 1 and 10
5. The probability of a certain event and of an impossible event are respectively
 - A. 1 and 1
 - B. 1 and 0
 - C. 0 and 0
 - D. 0 and 1
6. For any event E, $P(E) + P(\text{not } E) =$
 - A. 1
 - B. $\frac{1}{2}$
 - C. 0
 - D. 2

- 7.** The probability of an event is calculated as
- A. $\text{total} \div \text{favourable}$
 - B. $\text{favourable outcomes} \div \text{total outcomes}$
 - C. $\text{favourable} - \text{total}$
 - D. $\text{unfavourable} \div \text{favourable}$
- 8.** The fraction $\frac{1}{4}$ expressed as a decimal and as a percentage is
- A. 0.75 and 75%
 - B. 2.5 and 2.5%
 - C. 0.25 and 25%
 - D. 0.4 and 40%
- 9.** Drawing two bottle tops from a bag WITHOUT replacement gives
- A. dependent events
 - B. certain events
 - C. independent events
 - D. impossible events
- 10.** A bag has 3 red, 2 green and 1 pink bottle top. The probability that the first draw is red is
- A. $\frac{1}{3}$
 - B. $\frac{1}{6}$
 - C. $\frac{2}{3}$
 - D. $\frac{1}{2}$
- 11.** If the first draw (without replacement) is red, the probability that the second draw is red is
- A. $\frac{2}{5}$
 - B. $\frac{3}{5}$
 - C. $\frac{1}{6}$
 - D. $\frac{3}{6}$
- 12.** In the bottle-top experiment, does the first trial affect the second?
- A. Only when drawing with replacement
 - B. No — the bag stays the same
 - C. Only if a pink top is drawn
 - D. Yes — the first draw changes what is left in the bag
- 13.** Drawing two tops (without replacement) from 3 red, 2 green, 1 pink, the number of possible ordered colour pairs is
- A. 6
 - B. 8 (PP is impossible)
 - C. 9
 - D. 36

- 14.** Drawing WITH replacement gives
- A. mutually exclusive events
 - B. impossible events
 - C. dependent events
 - D. independent events
- 15.** A bag has 2 blue and 3 yellow marbles. Two are drawn without replacement. $P(\text{blue then blue}) =$
- A. $4/25$
 - B. $1/10$
 - C. $2/10$
 - D. $1/4$
- 16.** With the same bag, $P(\text{yellow then blue}) =$
- A. $3/20$
 - B. $6/25$
 - C. $3/10$
 - D. $1/10$
- 17.** The probability that the first card drawn from a full deck is an Ace is
- A. $1/13$
 - B. $4/13$
 - C. $1/52$
 - D. $1/4$
- 18.** The probability of drawing two Aces in two trials without replacement is
- A. $12/52$
 - B. $1/169$
 - C. $1/17$
 - D. $1/221$
- 19.** The probability of drawing an Ace and then a Jack, without replacement, is
- A. $16/2652 \times 2$
 - B. $1/221$
 - C. $4/663$
 - D. $8/52$
- 20.** After one Ace has been removed from a full deck, the probability the next card is an Ace is
- A. $3/52$
 - B. $4/51 \times 4$
 - C. $1/13$
 - D. $1/17$

- 21.** Drawing without replacement changes the probability of the second draw because the
- A. outcomes become equally likely
 - B. number of colours increases
 - C. total number of items left decreases
 - D. bag becomes heavier
- 22.** A bag has 4 red and 4 blue sweets. Two are taken without replacement. $P(\text{red then red}) =$
- A. $1/4$
 - B. $3/14$
 - C. $12/64$
 - D. $16/56$
- 23.** With the same bag, $P(\text{both sweets the same colour}) =$
- A. $1/2$
 - B. $4/7$
 - C. $3/7$
 - D. $3/14$
- 24.** For two dependent events A then B, $P(A \text{ and } B) =$
- A. $P(A) - P(B)$
 - B. $P(A) \times P(B \text{ given } A)$
 - C. $P(A) \times P(B)$ always
 - D. $P(A) + P(B)$
- 25.** $3/4$ expressed as a percentage is
- A. 34%
 - B. 43%
 - C. 75%
 - D. 80%
- 26.** 0.2 expressed as a fraction in lowest terms is
- A. $1/5$
 - B. $1/2$
 - C. $2/100$
 - D. $2/5$
- 27.** $1/8$ expressed as a decimal is
- A. 1.8
 - B. 0.8
 - C. 0.18
 - D. 0.125

- 28.** 40% expressed as a fraction in lowest terms is
- A. $\frac{4}{5}$
 - B. $\frac{2}{5}$
 - C. $\frac{40}{10}$
 - D. $\frac{1}{40}$
- 29.** A probability of $\frac{3}{5}$ means the ratio of favourable to unfavourable outcomes is
- A. 2 : 3
 - B. 5 : 3
 - C. 3 : 2
 - D. 3 : 5
- 30.** The probability of the path $R \rightarrow R$ is
- A. $\frac{6}{11}$
 - B. $\frac{3}{6}$
 - C. $\frac{2}{5}$
 - D. $\frac{1}{5}$
- 31.** The sum of the three first-stage branch probabilities ($\frac{3}{6} + \frac{2}{6} + \frac{1}{6}$) is
- A. 1
 - B. 0
 - C. 6
 - D. $\frac{1}{2}$
- 32.** $\frac{1}{221}$ expressed as a percentage to 2 significant figures is about
- A. 0.22%
 - B. 45%
 - C. 4.5%
 - D. 0.45%
- 33.** $\frac{1}{13}$ expressed as a percentage to 1 decimal place is about
- A. 0.77%
 - B. 7.7%
 - C. 7.5%
 - D. 13%
- 34.** A probability of $\frac{2}{5}$ gives the ratio favourable : unfavourable =
- A. 2 : 3
 - B. 2 : 5
 - C. 3 : 2
 - D. 5 : 2

- 35.** From the 3-red, 2-green, 1-pink bag, two tops drawn without replacement:
 $P(\text{green then green}) =$
- A. $1/9$
 - B. $4/36$
 - C. $2/15$
 - D. $1/15$
- 36.** From the same bag, $P(\text{at least one red in two draws}) =$
- A. $9/10$
 - B. $1/5$
 - C. $4/5$
 - D. $3/5$
- 37.** $1/10$ expressed as a percentage is
- A. 1%
 - B. 0.1%
 - C. 90%
 - D. 10%
- 38.** $3/10$ expressed as a decimal is
- A. 0.03
 - B. 0.3
 - C. 0.33
 - D. 3.0
- 39.** $4/663$ as a percentage is about
- A. 0.6%
 - B. 60%
 - C. 6%
 - D. 0.06%
- 40.** A probability of 0.45 expressed as a ratio favourable : unfavourable is
- A. 45 : 100
 - B. 9 : 20
 - C. 9 : 11
 - D. 11 : 9

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Sample space and dependence

Question 2

- (a) Tree diagram

Question 3

- (a) Card experiments (curriculum exemplars)

Question 4

- (a) Converting probabilities

Question 5

- (a) Dependent draws from 4 red and 4 blue

Question 6

- (a) At least one red

Chapter 10: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The number 38.4972 rounded to 2 decimal places is
 - A. 38.49
 - B. 38.497
 - C. 38.40
 - D. 38.50
2. The number 506,723,412 correct to 4 significant figures is
 - A. 507,000,000
 - B. 506,700,000
 - C. 506,723,400
 - D. 506,720,000
3. The number of significant figures in 70.030 is
 - A. 2
 - B. 3
 - C. 5
 - D. 4
4. The number 2,684,913 correct to 3 significant figures is
 - A. 2,680,000
 - B. 2,690,000
 - C. 2,684,000
 - D. 2,700,000
5. The number 9.996 rounded to 2 decimal places is
 - A. 9.96
 - B. 9.996
 - C. 10.00
 - D. 9.99
6. The number 0.0596 correct to 2 significant figures is
 - A. 0.0596
 - B. 0.060
 - C. 0.059
 - D. 0.06

- 7.** The number of significant figures in 0.004050 is
- A. 4
 - B. 3
 - C. 6
 - D. 7
- 8.** The number 74.996 rounded to 1 decimal place is
- A. 74.99
 - B. 75.9
 - C. 74.9
 - D. 75.0
- 9.** The numbers 0.501, 0.0501, 0.51 and 0.051 arranged in ascending order are
- A. 0.501, 0.51, 0.051, 0.0501
 - B. 0.0501, 0.051, 0.501, 0.51
 - C. 0.051, 0.0501, 0.501, 0.51
 - D. 0.0501, 0.501, 0.051, 0.51
- 10.** Which of the following numbers rounds to 6.0 correct to 1 decimal place?
- A. 5.94
 - B. 6.06
 - C. 5.9
 - D. 5.96
- 11.** The largest whole number which rounds to 45,000 to the nearest thousand is
- A. 44,500
 - B. 45,999
 - C. 45,499
 - D. 45,500
- 12.** The smallest number which rounds to 8.3 correct to 1 decimal place is
- A. 8.25
 - B. 8.31
 - C. 8.26
 - D. 8.24
- 13.** A whole number rounds to 700 correct to 2 significant figures. The largest possible value of the number is
- A. 750
 - B. 704
 - C. 799
 - D. 749

- 14.** By rounding each number to 1 significant figure, the estimated value of 62×19 is
- A. 600
 - B. 1,800
 - C. 1,200
 - D. 1,178
- 15.** I am a 5-digit number. My ten-thousands digit is 3. My thousands digit is twice my ten-thousands digit. My hundreds digit is 4 less than my thousands digit. My tens digit is 0 and my units digit is 9. The number is
- A. 36,209
 - B. 36,290
 - C. 36,609
 - D. 32,609
- 16.** The number in question 15 expressed in standard form is
- A. 3.6209×10^5
 - B. 3.6209×10^4
 - C. 36.209×10^3
 - D. 3.6209×10^3
- 17.** The place value of the digit 8 in the number 4.0805 is
- A. $\frac{8}{100}$
 - B. $\frac{8}{1,000}$
 - C. 8
 - D. $\frac{8}{10}$
- 18.** The value of the digit 5 in the number 2.0053 is
- A. 0.5
 - B. 0.05
 - C. 0.005
 - D. 0.0005
- 19.** The number 83,650,000 written in standard form is
- A. 8.365×10^6
 - B. 83.65×10^6
 - C. 8.365×10^8
 - D. 8.365×10^7
- 20.** Written as an ordinary number, 4.07×10^5 is
- A. 4,070,000
 - B. 407,000
 - C. 4,070
 - D. 40,700

- 21.** The number 0.0000743 written in standard form is
- A. 7.43×10^{-6}
 - B. 7.43×10^{-4}
 - C. 7.43×10^{-5}
 - D. 74.3×10^{-6}
- 22.** Written as a decimal, 2.05×10^{-3} is
- A. 0.00205
 - B. 0.0205
 - C. 0.000205
 - D. 0.205
- 23.** Evaluate, leaving your answer in standard form: $(5.6 \times 10^4) + (3.2 \times 10^4)$
- A. 88×10^4
 - B. 8.8×10^4
 - C. 8.8×10^3
 - D. 8.8×10^8
- 24.** Evaluate, leaving your answer in standard form: $(6.4 \times 10^6) \div (8 \times 10^2)$
- A. 8×10^2
 - B. 8×10^4
 - C. 0.8×10^4
 - D. 8×10^3
- 25.** The population of a region is 33,475,870. Correct to 2 significant figures, this is
- A. 33,500,000
 - B. 34,000,000
 - C. 33,000,000
 - D. 33,400,000
- 26.** A farmer harvested 127,483 kg of maize. Correct to 3 significant figures, the harvest is
- A. 127,000 kg
 - B. 127,500 kg
 - C. 120,000 kg
 - D. 128,000 kg
- 27.** Ama paid GH¢50.00 for items costing GH¢37.45. Her change, correct to 1 decimal place, is
- A. GH¢13.00
 - B. GH¢12.60
 - C. GH¢12.50
 - D. GH¢12.55

- 28.** The length of a wire is 0.004567 m. Correct to 2 significant figures, the length is
- A. 0.0045 m
 - B. 0.005 m
 - C. 0.00457 m
 - D. 0.0046 m
- 29.** The number of significant figures in 0.03080 is
- A. 5
 - B. 2
 - C. 4
 - D. 3
- 30.** Written as an ordinary number, 3.678×10^4 is
- A. 36,780
 - B. 36.78
 - C. 367,800
 - D. 3,678
- 31.** The number 504,673 rounded to the nearest ten thousand is
- A. 510,000
 - B. 500,000
 - C. 504,700
 - D. 505,000
- 32.** Evaluate, leaving your answer in standard form: $(2.4 \times 10^3) \times (3 \times 10^2)$
- A. 72×10^5
 - B. 5.4×10^5
 - C. 7.2×10^6
 - D. 7.2×10^5
- 33.** The distance between two towns is 8,345,000 m. Written in standard form, this is
- A. 8.345×10^7 m
 - B. 8.345×10^5 m
 - C. 8.345×10^6 m
 - D. 0.8345×10^7 m
- 34.** The population of a city is 3.2×10^6 . Written in full, the population is
- A. 32,000,000
 - B. 3,200,000
 - C. 320,000
 - D. 3,020,000

35. $\sqrt{63}$ lies between which two consecutive whole numbers?

- A. 7 and 8
- B. 8 and 9
- C. 31 and 32
- D. 6 and 7

36. The best estimate of $\sqrt{75}$, correct to 1 decimal place, is

- A. 9.4
- B. 7.5
- C. 8.0
- D. 8.7

37. Correct to 2 decimal places, $\sqrt{12}$ is

- A. 3.46
- B. 3.45
- C. 3.47
- D. 6.00

38. If $x^2 = 145$, the value of x correct to 1 decimal place is

- A. 11.9
- B. 72.5
- C. 12.0
- D. 12.1

39. Which of the following numbers is a perfect square?

- A. 99
- B. 20
- C. 50
- D. 81

40. Correct to 1 decimal place, $\sqrt{101}$ is

- A. 10.5
- B. 10.0
- C. 9.5
- D. 10.1

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Rounding numbers.

Question 2

(a) Standard form.

Question 3

(a) Place value puzzle.

Question 4

(a) Upper and lower bounds.

Question 5

(a) Approximating square roots.

Question 6

(a) Integrated problems.

Chapter 11: Ch2

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following numbers is irrational?
 - A. 0.75
 - B. $\frac{2}{3}$
 - C. $\sqrt{2}$
 - D. $\sqrt{49}$
2. The set of natural numbers is usually denoted by
 - A. Q
 - B. W
 - C. Z
 - D. N
3. Every integer is also a
 - A. rational number
 - B. whole number
 - C. irrational number
 - D. natural number
4. Which of the following is a whole number but NOT a natural number?
 - A. 1
 - B. 0
 - C. -1
 - D. $\frac{1}{2}$
5. The decimal expansion of $\frac{22}{7}$ is
 - A. non-terminating and non-repeating
 - B. irrational
 - C. terminating
 - D. non-terminating and repeating
6. $\sqrt{49}$ simplifies to 7, so it is
 - A. a prime number
 - B. a rational number
 - C. an irrational number
 - D. not a real number

7. Which of the following best describes the relationship between the set of natural numbers N and the set of integers Z ?
- A. $N \subset Z$
 - B. $Z \subset N$
 - C. $N = Z$
 - D. $N \cap Z =$
8. The intersection of the set of rational numbers and the set of irrational numbers is
- A. the set of real numbers
 - B. the set of rational numbers
 - C. the empty set,
 - D. $\{0\}$
9. The union of the set of rational numbers and the set of irrational numbers gives
- A. the set of whole numbers
 - B. the set of integers
 - C. the set of rational numbers
 - D. the set of real numbers
10. Which of the following is NOT a rational number?
- A. $0.333\dots$
 - B. $\sqrt{5}$
 - C. -8
 - D. $3/4$
11. π is best classified as
- A. a whole number
 - B. an integer
 - C. an irrational number
 - D. a rational number
12. Which of the following numbers belongs to Z but NOT to W ?
- A. -6
 - B. 0.6
 - C. 0
 - D. 6
13. If $A = \{\text{factors of } 12\}$ and $B = \{\text{factors of } 15\}$, then $A \cap B =$
- A. $\{1, 2, 3, 4, 6, 12\}$
 - B. $\{3, 5, 15\}$
 - C. $\{1, 3\}$
 - D. $\{1\}$

- 14.** From the sets in question 13, $A \cup B =$
- A. $\{1, 2, 3, 4, 6, 12, 15\}$
 - B. $\{2, 4, 5, 6, 12, 15\}$
 - C. $\{1, 3\}$
 - D. $\{1, 2, 3, 4, 5, 6, 12, 15\}$
- 15.** The number of elements in $A \cup B$, for the sets in question 13, is
- A. 8
 - B. 2
 - C. 10
 - D. 6
- 16.** In a Venn diagram, the overlapping region of two circles represents the
- A. union of the sets
 - B. intersection of the sets
 - C. complement of a set
 - D. universal set
- 17.** The set that contains all the elements under consideration is called the
- A. subset
 - B. complement
 - C. universal set
 - D. empty set
- 18.** If $n(A) = 6$, $n(B) = 4$ and $n(A \cap B) = 2$, then $n(A \cup B) =$
- A. 24
 - B. 8
 - C. 10
 - D. 12
- 19.** A set which has no elements is called the
- A. universal set
 - B. unit set
 - C. subset
 - D. empty set
- 20.** If $P = \{2, 4, 6\}$ and $Q = \{4, 6, 8\}$, then $P \cap Q =$
- A. $\{4, 6\}$
 - B. $\{2, 4, 6, 8\}$
 - C. $\{2, 4, 6\}$
 - D. $\{2, 8\}$

- 21.** In a class of 40 pupils, 25 like Mathematics and 20 like Science. If every pupil likes at least one of the two subjects, the number who like both subjects is
- A. 45
 - B. 15
 - C. 5
 - D. 20
- 22.** From the information in question 21, the number of pupils who like Mathematics only is
- A. 25
 - B. 20
 - C. 15
 - D. 5
- 23.** From the information in question 21, the number of pupils who like exactly one subject is
- A. 25
 - B. 15
 - C. 40
 - D. 35
- 24.** 100 people were surveyed: 60 drink tea, 50 drink coffee and 10 drink neither. The number who drink both tea and coffee is
- A. 20
 - B. 110
 - C. 10
 - D. 30
- 25.** From the survey in question 24, the number of people who drink coffee only is
- A. 50
 - B. 20
 - C. 30
 - D. 40
- 26.** From the survey in question 24, the probability that a person chosen at random drinks tea only is
- A. $\frac{1}{2}$
 - B. $\frac{2}{5}$
 - C. $\frac{3}{10}$
 - D. $\frac{3}{5}$

- 27.** In a village of 80 farmers, 50 grow beans and 60 grow maize. If each farmer grows at least one of the two crops, the number who grow both crops is
- A. 10
 - B. 20
 - C. 50
 - D. 30
- 28.** From the information in question 27, the number of farmers who grow only one crop is
- A. 50
 - B. 30
 - C. 20
 - D. 80
- 29.** In a group of 60 people, 35 speak English, 30 speak French and 12 speak both languages. The number who speak neither language is
- A. 13
 - B. 17
 - C. 12
 - D. 7
- 30.** From the information in question 29, the number of people who speak exactly one of the two languages is
- A. 18
 - B. 41
 - C. 53
 - D. 35
- 31.** In a survey of 100 students, 55 study Mathematics, 45 study Science and 40 study English; 25 study Mathematics and Science, 20 study Science and English, 15 study Mathematics and English, and 10 study all three subjects. The number who study Mathematics only is
- A. 55
 - B. 15
 - C. 25
 - D. 35
- 32.** From the survey in question 31, the number of students who study at least one of the three subjects is
- A. 90
 - B. 80
 - C. 75
 - D. 100

- 33.** From the survey in question 31, the number of students who study none of the three subjects is
- A. 90
 - B. 10
 - C. 20
 - D. 0
- 34.** From the survey in question 31, the number of students who study exactly two subjects is
- A. 60
 - B. 50
 - C. 30
 - D. 20
- 35.** In a survey of 120 people about drinks, 60 take tea, 50 take coffee and 40 take Milo; 20 take tea and coffee, 15 take coffee and Milo, 10 take tea and Milo, and 5 take all three. The number who take tea and coffee but not Milo is
- A. 10
 - B. 25
 - C. 20
 - D. 15
- 36.** From the survey in question 35, the number of people who take only Milo is
- A. 20
 - B. 25
 - C. 40
 - D. 15
- 37.** From the survey in question 35, the number of people who take at least one of the drinks is
- A. 120
 - B. 100
 - C. 110
 - D. 105
- 38.** From the survey in question 35, a person is chosen at random. The probability that the person takes none of the three drinks is
- A. $\frac{11}{12}$
 - B. $\frac{1}{12}$
 - C. $\frac{1}{6}$
 - D. $\frac{1}{10}$

39. A Venn diagram for three sets divides the universal set into how many regions in all?

- A. 8
- B. 3
- C. 6
- D. 7

40. For any three sets A, B and C, $n(A \cup B \cup C)$ is equal to

- A. $n(A) + n(B) + n(C) - n(A \cap B \cap C)$
- B. $n(A) + n(B) + n(C)$
- C. $n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C)$
- D. $n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) The number system.

Question 2

- (a) Sets of factors.

Question 3

- (a) A two-set survey.

Question 4

- (a) A three-set survey.

Question 5

- (a) Three sets with an unknown.

Question 6

- (a) Design your own survey.

Chapter 12: Ch3

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. 4.73×100 is equal to
 - A. 47.3
 - B. 473
 - C. 4,730
 - D. 0.0473
2. $85 \div 1,000$ is equal to
 - A. 8.5
 - B. 0.0085
 - C. 0.085
 - D. 0.85
3. 0.09×10^3 is equal to
 - A. 90
 - B. 0.9
 - C. 9
 - D. 900
4. $620 \div 10^2$ is equal to
 - A. 62
 - B. 0.62
 - C. 6,200
 - D. 6.2
5. The benchmark fraction equivalent to 0.25 is
 - A. $\frac{2}{5}$
 - B. $\frac{1}{4}$
 - C. $\frac{1}{2}$
 - D. $\frac{1}{5}$
6. $\frac{3}{5}$ written as a decimal is
 - A. 0.6
 - B. 0.53
 - C. 0.35
 - D. 3.5

- 7.** $12.5 \times 1,000$ is equal to
- A. 1,250
 - B. 125,000
 - C. 125
 - D. 12,500
- 8.** $7 \div 100$ is equal to
- A. 70
 - B. 0.007
 - C. 0.07
 - D. 0.7
- 9.** 0.375 written as a benchmark fraction is
- A. $\frac{1}{3}$
 - B. $\frac{3}{4}$
 - C. $\frac{5}{8}$
 - D. $\frac{3}{8}$
- 10.** Multiplying a number by 10^{-2} has the same effect as
- A. dividing the number by 100
 - B. multiplying the number by 100
 - C. dividing the number by 2
 - D. multiplying the number by 2
- 11.** Which of the following illustrates the commutative property of addition?
- A. $8 + 0 = 8$
 - B. $8 \times 1 = 8$
 - C. $8 + 3 = 3 + 8$
 - D. $(8 + 3) + 2 = 8 + (3 + 2)$
- 12.** $17 \times 8 = 8 \times 17$ is equal to
- A. 153
 - B. 136
 - C. 126
 - D. 146
- 13.** Which of the following operations is NOT commutative?
- A. Addition
 - B. Multiplication
 - C. Both addition and multiplication
 - D. Subtraction

- 14.** If $23 + 48 = 48 + 23$, the property illustrated is the commutative property of
- A. subtraction
 - B. division
 - C. addition
 - D. multiplication
- 15.** A student wrote $7 \times 9 = 9 \times 7$. The property used is the
- A. commutative property of addition
 - B. commutative property of multiplication
 - C. associative property of multiplication
 - D. distributive property
- 16.** By the commutative property of multiplication, $a \times b =$
- A. $b \times a$
 - B. $a + b$
 - C. $a \div b$
 - D. $b \div a$
- 17.** $(8 + 14) + 22 = 8 + (14 + 22)$ illustrates the associative property of
- A. commutative property of addition
 - B. addition
 - C. distributive property
 - D. multiplication
- 18.** $(5 \times 6) \times 2 = 5 \times (6 \times 2)$ is equal to
- A. 60
 - B. 30
 - C. 50
 - D. 62
- 19.** Which of the following illustrates the associative property of multiplication?
- A. $2 \times 3 = 3 \times 2$
 - B. $2 \times (3 + 5) = (2 \times 3) + (2 \times 5)$
 - C. $(2 \times 3) \times 5 = 2 \times (3 \times 5)$
 - D. $2 \times 5 = 5 \times 2$
- 20.** $(15 + 6) + 9 = 15 + (6 + 9)$ is equal to
- A. 39
 - B. 24
 - C. 21
 - D. 30

- 21.** By the associative property of multiplication, $(a \times b) \times c =$
- A. $a \times (b \times c)$
 - B. $(a + b) \times c$
 - C. $(a \times b) + c$
 - D. $a + (b \times c)$
- 22.** Using the associative property, the easiest way to evaluate $(25 \times 17) \times 4$ gives
- A. 1,070
 - B. 170
 - C. 1,700
 - D. 1,600
- 23.** $5 \times (10 + 7) = (5 \times 10) + (5 \times 7)$ is equal to
- A. 120
 - B. 75
 - C. 57
 - D. 85
- 24.** $5 \times (10 - 7) = (5 \times 10) - (5 \times 7)$ is equal to
- A. 50
 - B. 15
 - C. 35
 - D. 53
- 25.** Using the distributive property, $6 \times 28 = 6 \times (30 - 2)$ is equal to
- A. 158
 - B. 178
 - C. 162
 - D. 168
- 26.** $7 \times 46 = (7 \times 40) + (7 \times 6)$ is equal to
- A. 286
 - B. 332
 - C. 322
 - D. 316
- 27.** Using the distributive property, $13 \times 99 = 13 \times (100 - 1)$ is equal to
- A. 1,313
 - B. 1,287
 - C. 1,297
 - D. 1,277

- 28.** $6 \times (20 + 3) = (6 \times 20) + (6 \times 3)$ is equal to
- A. 138
 - B. 123
 - C. 128
 - D. 148
- 29.** Which of the following states the distributive property?
- A. $a + b = b + a$
 - B. $a \times (b + c) = (a \times b) + (a \times c)$
 - C. $a \times (b + c) = (a + b) \times (a + c)$
 - D. $(a \times b) \times c = a \times (b \times c)$
- 30.** Using the distributive property, $9 \times (50 - 4)$ is equal to
- A. 414
 - B. 450
 - C. 404
 - D. 424
- 31.** Mentally, 0.5×36 is equal to
- A. 17
 - B. 19
 - C. 16
 - D. 18
- 32.** $15 \times 12 + 15 \times 8 = 15 \times (12 + 8)$ is equal to
- A. 280
 - B. 315
 - C. 300
 - D. 200
- 33.** Using a mental strategy, 48×25 is equal to
- A. 1,100
 - B. 1,200
 - C. 1,080
 - D. 1,400
- 34.** Using compensation, $49 + 38$ is equal to
- A. 77
 - B. 97
 - C. 87
 - D. 88

- 35.** Using compensation, $132 - 98$ is equal to
- A. 34
 - B. 24
 - C. 36
 - D. 44
- 36.** Mentally, 76×5 is equal to
- A. 350
 - B. 385
 - C. 760
 - D. 380
- 37.** By grouping to make tens, $19 + 27 + 31 + 23$ is equal to
- A. 100
 - B. 90
 - C. 110
 - D. 99
- 38.** If $16 \times 7 = 112$, then 16×14 is equal to
- A. 212
 - B. 252
 - C. 126
 - D. 224
- 39.** Using a mental strategy, 250×16 is equal to
- A. 3,600
 - B. 2,500
 - C. 4,000
 - D. 4,500
- 40.** Amina buys 8 exercise books at GH¢4.50 each and pays with a GH¢50 note. Her change is
- A. GH¢4.00
 - B. GH¢14.00
 - C. GH¢24.00
 - D. GH¢8.50

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Powers of 10 and benchmark fractions.

Question 2

- (a) The commutative and associative properties.

Question 3

- (a) The distributive property.

Question 4

- (a) Mental mathematics strategies.

Question 5

- (a) Applications.

Question 6

- (a) Reasoning with properties.

Chapter 13: Ch

Time Allowed: 2 Hours | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Evaluate: $4.6 + 3.75$
 - A. 7.81
 - B. 8.21
 - C. 4.975
 - D. 8.35
2. Evaluate: $9.2 - 3.86$
 - A. 5.34
 - B. 5.44
 - C. 6.34
 - D. 6.66
3. Evaluate: $0.58 + 0.047$
 - A. 0.99
 - B. 0.105
 - C. 0.627
 - D. 0.605
4. Evaluate: $7 - 2.46$
 - A. 5.54
 - B. 4.54
 - C. 4.46
 - D. 5.46
5. Evaluate: $12.35 + 8.9 + 3.06$
 - A. 24.41
 - B. 25.31
 - C. 23.31
 - D. 24.31
6. Evaluate: $15.06 - 7.48$
 - A. 8.58
 - B. 8.42
 - C. 7.58
 - D. 7.42

7. Evaluate: $0.9 - 0.009$

- A. 0.899
- B. 0.891
- C. 0.909
- D. 0.9

8. Evaluate: $3.75 + 2.5 + 0.25$

- A. 6.5
- B. 6.55
- C. 8.75
- D. 6.05

9. Evaluate: $10.005 - 4.7$

- A. 5.295
- B. 5.705
- C. 5.305
- D. 6.295

10. Ama spent GH¢18.45 on food and GH¢6.75 on transport. The total amount she spent is

- A. GH¢24.20
- B. GH¢25.20
- C. GH¢25.02
- D. GH¢24.85

11. Evaluate: 3.6×7

- A. 252
- B. 24.2
- C. 2.52
- D. 25.2

12. Evaluate: 0.25×0.4

- A. 0.1
- B. 0.65
- C. 1.0
- D. 0.01

13. Evaluate: $8.4 \div 4$

- A. 21
- B. 2.1
- C. 0.21
- D. 2.4

14. Evaluate: $6.3 \div 0.9$

- A. 70
- B. 5.67
- C. 0.7
- D. 7

15. Evaluate: 2.5×2.5

- A. 5.25
- B. 5.0
- C. 6.25
- D. 62.5

16. Evaluate: $9.36 \div 12$

- A. 0.78
- B. 7.8
- C. 0.87
- D. 0.078

17. Evaluate: 0.12×0.3

- A. 3.6
- B. 0.036
- C. 0.0036
- D. 0.36

18. Evaluate: $4.8 \div 0.6$

- A. 2.88
- B. 0.8
- C. 8
- D. 80

19. Evaluate: 7.2×0.05

- A. 3.6
- B. 0.036
- C. 36
- D. 0.36

20. Evaluate: $14.4 \div 1.2$

- A. 12
- B. 120
- C. 13.2
- D. 1.2

- 21.** The value of 3.6×4.7 , correct to 1 decimal place, is
- A. 16.8
 - B. 17.0
 - C. 16.92
 - D. 16.9
- 22.** The value of $19.87 \div 6$, correct to 2 decimal places, is
- A. 3.32
 - B. 3.30
 - C. 3.31
 - D. 3.312
- 23.** The value of 5.678×1.4 , correct to 2 decimal places, is
- A. 7.95
 - B. 7.949
 - C. 8.05
 - D. 7.94
- 24.** The value of $25.6 \div 7$, correct to 3 significant figures, is
- A. 3.7
 - B. 3.66
 - C. 3.65
 - D. 3.657
- 25.** The value of $8.91 - 2.348$, correct to 2 decimal places, is
- A. 6.57
 - B. 6.562
 - C. 6.56
 - D. 6.66
- 26.** The value of $(4.6 \times 7) \div 9$, correct to 2 decimal places, is
- A. 3.577
 - B. 3.58
 - C. 3.6
 - D. 3.57
- 27.** The value of $23.2 \div 3.4$, correct to 2 decimal places, is
- A. 6.72
 - B. 6.83
 - C. 6.824
 - D. 6.82

- 28.** A rope 12.5 m long is cut into 6 equal pieces. The length of each piece, correct to 2 decimal places, is
- A. 2.08 m
 - B. 2.09 m
 - C. 2.083 m
 - D. 2.18 m
- 29.** A trader sells oranges from two baskets. Basket A contained 85 oranges and she sold 48 of them. The number of oranges left in basket A is
- A. 133
 - B. 38
 - C. 37
 - D. 47
- 30.** From the information in question 29: the trader sold 59 oranges from basket B and was left with the same number as remained in basket A. The number of oranges originally in basket B was
- A. 37
 - B. 96
 - C. 86
 - D. 106
- 31.** Kofi bought rice for GH¢145.75 and cooking oil for GH¢68.40. The total cost of the items is
- A. GH¢213.15
 - B. GH¢204.15
 - C. GH¢214.25
 - D. GH¢214.15
- 32.** From the information in question 31, if Kofi paid with GH¢300, his change is
- A. GH¢85.85
 - B. GH¢85.75
 - C. GH¢86.15
 - D. GH¢95.85
- 33.** One bottle holds 0.75 litre of juice. The total volume of juice in 8 such bottles is
- A. 60 litres
 - B. 6 litres
 - C. 5.6 litres
 - D. 6.75 litres

- 34.** 9.6 kg of sugar is shared equally into 4 bags. The mass of sugar in each bag is
- A. 2.04 kg
 - B. 24 kg
 - C. 2.4 kg
 - D. 3.6 kg
- 35.** Adjoa runs 2.35 km every morning. The total distance she runs in 5 days is
- A. 11.75 km
 - B. 12.25 km
 - C. 11.35 km
 - D. 10.75 km
- 36.** From the information in question 35, the total distance Adjoa runs in one week (7 days) is
- A. 17.45 km
 - B. 16.10 km
 - C. 15.45 km
 - D. 16.45 km
- 37.** A rectangle has length 8.4 cm and width 3.25 cm. Its perimeter is
- A. 23.3 cm
 - B. 27.3 cm
 - C. 11.65 cm
 - D. 22.9 cm
- 38.** The area of the rectangle in question 37 is
- A. 27.03 cm²
 - B. 11.65 cm²
 - C. 27.3 cm²
 - D. 23.3 cm²
- 39.** A minibus used 12.8 litres of fuel over 4 days, the same amount each day. The fuel used per day is
- A. 4.8 litres
 - B. 3.02 litres
 - C. 32 litres
 - D. 3.2 litres
- 40.** Ama has a mass of 42.6 kg and her younger brother has a mass of 38.85 kg. The difference in their masses is
- A. 4.25 kg
 - B. 3.75 kg
 - C. 3.25 kg
 - D. 4.75 kg

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Adding and subtracting decimals.

Question 2

(a) Multiplying and dividing decimals.

Question 3

(a) Mixed operations (use the order of operations).

Question 4

(a) The egg-seller's problem.

Question 5

(a) Shopping with rounding.

Question 6

(a) Create and solve.

Chapter 14: Ch5

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following is a surd?
 - A. $\sqrt{4}$
 - B. $\sqrt{16}$
 - C. $\sqrt{2}$
 - D. $\sqrt{36}$
2. Evaluate $\sqrt{16}$.
 - A. 2
 - B. 36
 - C. $\sqrt{8}$
 - D. 4
3. Which of the following is a compound surd?
 - A. $\sqrt{2}$
 - B. $2\sqrt{5}$
 - C. $\sqrt{3} + \sqrt{7} - \sqrt{5}$
 - D. $7\sqrt{3}$
4. Which of the following surds are like surds?
 - A. $\sqrt{2}$ and $\sqrt{3}$
 - B. $2\sqrt{5}$ and $\sqrt{45}$
 - C. $\sqrt{3}$ and $\sqrt{27}$
 - D. $\sqrt{8}$ and $\sqrt{18}$
5. The square root of a number which is not a perfect square is
 - A. a surd
 - B. always a fraction
 - C. an integer
 - D. a whole number
6. Which of the following is NOT a surd?
 - A. $\sqrt{11}$
 - B. $\sqrt{49}$
 - C. $\sqrt{5}$
 - D. $\sqrt{7}$

7. Simplify $\sqrt{8}$.

- A. $2\sqrt{2}$
- B. $4\sqrt{2}$
- C. 8
- D. $2\sqrt{3}$

8. Simplify $\sqrt{12}$.

- A. 6
- B. $4\sqrt{3}$
- C. $3\sqrt{2}$
- D. $2\sqrt{3}$

9. Simplify $\sqrt{20}$.

- A. $5\sqrt{2}$
- B. $2\sqrt{5}$
- C. 10
- D. $4\sqrt{5}$

10. Simplify $\sqrt{27}$.

- A. $9\sqrt{3}$
- B. $3\sqrt{9}$
- C. 27
- D. $3\sqrt{3}$

11. Simplify $\sqrt{45}$.

- A. $3\sqrt{5}$
- B. $9\sqrt{5}$
- C. 5
- D. $5\sqrt{3}$

12. Simplify $\sqrt{48}$.

- A. $2\sqrt{6}$
- B. 24
- C. $6\sqrt{2}$
- D. $4\sqrt{3}$

13. Simplify $\sqrt{50}$.

- A. $10\sqrt{5}$
- B. $25\sqrt{2}$
- C. $5\sqrt{2}$
- D. $2\sqrt{5}$

14. Simplify $\sqrt{72}$.

- A. $2\sqrt{6}$
- B. $8\sqrt{3}$
- C. 36
- D. $6\sqrt{2}$

15. Simplify $\sqrt{75}$.

- A. 15
- B. $3\sqrt{5}$
- C. $5\sqrt{3}$
- D. $25\sqrt{3}$

16. Simplify $\sqrt{200}$.

- A. $10\sqrt{2}$
- B. 100
- C. 20
- D. $2\sqrt{10}$

17. Simplify $3\sqrt{2} + 5\sqrt{2}$.

- A. $8\sqrt{10}$
- B. 35
- C. $8\sqrt{2}$
- D. $15\sqrt{2}$

18. Simplify $9\sqrt{5} - 4\sqrt{5}$.

- A. $5\sqrt{5}$
- B. 5
- C. $5\sqrt{10}$
- D. $\sqrt{5}$

19. Simplify $7\sqrt{2} + 3\sqrt{2}$.

- A. $10\sqrt{10}$
- B. $10\sqrt{2}$
- C. 21
- D. 10

20. Simplify $6\sqrt{3} + 4\sqrt{3}$.

- A. $\sqrt{3}$
- B. $10\sqrt{9}$
- C. $10\sqrt{3}$
- D. 10

21. Simplify $4\sqrt{7} + 5\sqrt{7}$.

- A. $\sqrt{7}$
- B. $9\sqrt{7}$
- C. $9\sqrt{35}$
- D. $20\sqrt{7}$

22. Simplify $3\sqrt{5} + 2\sqrt{5} - \sqrt{5}$.

- A. $6\sqrt{5}$
- B. $\sqrt{5}$
- C. $4\sqrt{5}$
- D. 0

23. Simplify $7\sqrt{2} - 2\sqrt{2}$.

- A. $\sqrt{2}$
- B. $5\sqrt{2}$
- C. 14
- D. $9\sqrt{2}$

24. Which of the following is equal to $5\sqrt{2}$?

- A. $\sqrt{45}$
- B. $\sqrt{20}$
- C. $\sqrt{12}$
- D. $\sqrt{50}$

25. Simplify $\sqrt{18} + \sqrt{8}$.

- A. $5\sqrt{2}$
- B. $\sqrt{5}$
- C. 10
- D. $5\sqrt{10}$

26. Simplify $3\sqrt{2} + 2\sqrt{2}$.

- A. $\sqrt{2}$
- B. 12
- C. $5\sqrt{2}$
- D. $6\sqrt{2}$

27. Simplify $3\sqrt{3} + 4\sqrt{3}$.

- A. $\sqrt{3}$
- B. $7\sqrt{3}$
- C. $12\sqrt{3}$
- D. 7

28. Simplify $2\sqrt{2} + 5\sqrt{2}$.

- A. 10
- B. $\sqrt{10}$
- C. $7\sqrt{20}$
- D. $7\sqrt{2}$

29. Simplify $\sqrt{12} + \sqrt{27}$.

- A. $7\sqrt{5}$
- B. $\sqrt{15}$
- C. $5\sqrt{3}$
- D. $5\sqrt{15}$

30. Simplify $\sqrt{3} + \sqrt{5}$.

- A. $2\sqrt{2}$
- B. $\sqrt{8}$
- C. $\sqrt{15}$
- D. $\sqrt{3} + \sqrt{5}$

31. Evaluate $\sqrt{2} \times \sqrt{3}$.

- A. $\sqrt{6}$
- B. $\sqrt{5}$
- C. 6
- D. $\sqrt{23}$

32. Evaluate $\sqrt{5} \times \sqrt{5}$.

- A. 25
- B. 10
- C. $\sqrt{10}$
- D. 5

33. Evaluate $2\sqrt{3} \times \sqrt{2}$.

- A. $2\sqrt{6}$
- B. 12
- C. $6\sqrt{6}$
- D. $\sqrt{6}$

34. Evaluate $3\sqrt{5} \times 2\sqrt{5}$.

- A. $6\sqrt{5}$
- B. 30
- C. 15
- D. $30\sqrt{5}$

35. Evaluate $\sqrt{8} \times \sqrt{2}$.

- A. 16
- B. 8
- C. 2
- D. 4

36. Evaluate $\sqrt{7} \times \sqrt{28}$.

- A. 98
- B. 14
- C. $\sqrt{35}$
- D. 7

37. Simplify $\sqrt{12} \div \sqrt{3}$.

- A. 4
- B. $\sqrt{9}$
- C. 2
- D. 12

38. Simplify $\sqrt{50} \div \sqrt{2}$.

- A. 5
- B. $\sqrt{48}$
- C. 10
- D. 25

39. Simplify $\sqrt{18} \div \sqrt{2}$.

- A. $\sqrt{9}$
- B. 3
- C. 9
- D. $\sqrt{16}$

40. Simplify $(6\sqrt{2}) \div (3\sqrt{2})$.

- A. 2
- B. 3
- C. $2\sqrt{2}$
- D. 6

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Simplifying surds

Question 2

- (a) Adding and subtracting surds

Question 3

- (a) Multiplying and dividing surds

Question 4

- (a) Surds in context

Question 5

- (a) Identities and exact values

Question 6

- (a) Create and solve

Chapter 15: Ch

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Which of the following fractions is equivalent to $\frac{2}{5}$?
 - A. $\frac{3}{10}$
 - B. $\frac{5}{2}$
 - C. $\frac{2}{10}$
 - D. $\frac{4}{10}$
2. Express $\frac{15}{10}$ in its simplest form.
 - A. $\frac{3}{2}$
 - B. $\frac{1}{2}$
 - C. $\frac{5}{10}$
 - D. $\frac{5}{3}$
3. Express $\frac{12}{5}$ as a mixed number.
 - A. $2 \frac{3}{5}$
 - B. $2 \frac{1}{5}$
 - C. $1 \frac{2}{5}$
 - D. $2 \frac{2}{5}$
4. Express $2 \frac{5}{9}$ as an improper fraction.
 - A. $\frac{23}{9}$
 - B. $\frac{19}{9}$
 - C. $\frac{14}{9}$
 - D. $\frac{25}{9}$
5. Which of the following fractions is the greatest?
 - A. $\frac{5}{8}$
 - B. $\frac{3}{4}$
 - C. $\frac{7}{12}$
 - D. $\frac{2}{3}$
6. Which of the following fractions lies between $\frac{1}{4}$ and $\frac{1}{2}$?
 - A. $\frac{3}{4}$
 - B. $\frac{1}{8}$
 - C. $\frac{5}{8}$
 - D. $\frac{3}{8}$

- 7.** Reduce $18/24$ to its lowest terms.
- A. $2/3$
 - B. $9/16$
 - C. $3/4$
 - D. $4/3$
- 8.** Express 0.75 as a fraction in its lowest terms.
- A. $3/5$
 - B. $3/4$
 - C. $7/25$
 - D. $7/10$
- 9.** Work out $3/4 + 7/8$.
- A. $1\ 5/8$
 - B. $1\ 3/8$
 - C. $10/12$
 - D. $1\ 1/8$
- 10.** Work out $1/2 + 1/3$.
- A. $2/5$
 - B. $1/6$
 - C. $2/6$
 - D. $5/6$
- 11.** Work out $5/6 - 1/4$.
- A. $2/3$
 - B. $1/12$
 - C. $7/12$
 - D. $1/3$
- 12.** Work out $1\ 1/2 + 4/5 - 5/6$.
- A. $1\ 4/5$
 - B. $1\ 7/15$
 - C. $1\ 2/15$
 - D. $11/15$
- 13.** Work out $2\ 1/4 - 1\ 1/2$.
- A. $1\ 1/4$
 - B. $1/2$
 - C. $3/4$
 - D. $1\ 3/4$

14. Work out $\frac{3}{8} + \frac{1}{6}$.

- A. $\frac{13}{24}$
- B. $\frac{5}{24}$
- C. $\frac{11}{24}$
- D. $\frac{4}{14}$

15. Work out $\frac{7}{9} - \frac{1}{6}$.

- A. $\frac{5}{18}$
- B. $\frac{1}{3}$
- C. $\frac{11}{18}$
- D. $\frac{1}{2}$

16. What must be added to $\frac{3}{7}$ to give 1?

- A. $\frac{3}{7}$
- B. $\frac{4}{7}$
- C. $\frac{7}{3}$
- D. $\frac{1}{7}$

17. Evaluate $\frac{2}{3} \times \frac{3}{4}$.

- A. $\frac{8}{9}$
- B. $\frac{1}{4}$
- C. $\frac{1}{2}$
- D. $\frac{5}{7}$

18. Evaluate $\frac{5}{6} \div \frac{2}{3}$.

- A. $\frac{3}{10}$
- B. $\frac{5}{9}$
- C. $\frac{4}{5}$
- D. $1 \frac{1}{4}$

19. Find $\frac{3}{5}$ of 40.

- A. 15
- B. 24
- C. 32
- D. 8

20. Evaluate $\frac{4}{9} \times \frac{3}{8}$.

- A. $\frac{1}{6}$
- B. $\frac{1}{2}$
- C. $\frac{2}{3}$
- D. $\frac{7}{17}$

21. Evaluate $2\frac{1}{2} \div \frac{1}{4}$.

- A. 4
- B. $\frac{5}{8}$
- C. 10
- D. $\frac{9}{8}$

22. Evaluate $(\frac{3}{4})^2$.

- A. $\frac{3}{2}$
- B. $\frac{9}{16}$
- C. $\frac{3}{8}$
- D. $\frac{9}{8}$

23. Evaluate $\frac{7}{8} \div 7$.

- A. $\frac{1}{56}$
- B. 8
- C. $\frac{49}{8}$
- D. $\frac{1}{8}$

24. Evaluate $\frac{5}{8} \div 2\frac{1}{2} + \frac{2}{3}$.

- A. $\frac{11}{12}$
- B. $\frac{1}{6}$
- C. $\frac{1}{4}$
- D. $\frac{13}{12}$

25. Using BODMAS, evaluate $\frac{2}{3} \times \frac{3}{4} - \frac{5}{8} \div 2\frac{1}{2}$.

- A. $\frac{1}{8}$
- B. $\frac{1}{4}$
- C. $\frac{3}{8}$
- D. $\frac{1}{2}$

26. Using BODMAS, evaluate $\frac{3}{4} \div \frac{3}{8} + (\frac{4}{5} - \frac{1}{2})$.

- A. $\frac{8}{5}$
- B. $1\frac{3}{10}$
- C. $2\frac{3}{10}$
- D. $2\frac{1}{10}$

27. Using BODMAS, evaluate $(\frac{3}{4} + \frac{5}{8}) \times \frac{4}{11} - \frac{1}{2}$.

- A. 2
- B. 1
- C. $\frac{1}{2}$
- D. 0

- 28.** Evaluate $\frac{1}{2} + \frac{1}{4} \times \frac{2}{3}$.
- A. $\frac{1}{2}$
 - B. $\frac{1}{4}$
 - C. $\frac{2}{3}$
 - D. $\frac{5}{6}$
- 29.** Evaluate $(\frac{2}{3} + \frac{1}{6}) \div \frac{5}{12}$.
- A. 2
 - B. $\frac{5}{2}$
 - C. $\frac{1}{2}$
 - D. 1
- 30.** Evaluate $\frac{3}{5} \times (\frac{1}{2} + \frac{1}{3})$.
- A. $\frac{5}{6}$
 - B. $\frac{1}{2}$
 - C. $\frac{3}{10}$
 - D. $\frac{1}{5}$
- 31.** Evaluate $4 - 1\frac{1}{2} \div \frac{3}{4}$.
- A. $\frac{10}{3}$
 - B. 1
 - C. 3
 - D. 2
- 32.** Find $\frac{2}{3}$ of $\frac{1}{4}$ of 60.
- A. 20
 - B. 5
 - C. 10
 - D. 40
- 33.** A test is made up of 20 questions. How many questions must you answer correctly to get a score of 80%?
- A. 16
 - B. 18
 - C. 12
 - D. 8
- 34.** A television set was marked GH¢2,250 and sold for GH¢2,025. By what percentage was it reduced?
- A. 90%
 - B. 10%
 - C. 225%
 - D. 25%

- 35.** In an election, 5,000 people voted. One candidate claimed 52% of the votes while the other claimed 2,681 votes. What is the maximum number of votes the second candidate could actually have received?
- A. 2,681
 - B. 2,500
 - C. 2,400
 - D. 2,600
- 36.** Esi's drink is made of $\frac{3}{8}$ orange squash and Fusena's of $\frac{2}{5}$ orange squash. Whose drink tastes stronger of orange?
- A. Fusena's drink
 - B. Cannot be determined
 - C. Esi's drink
 - D. Both taste equally strong
- 37.** Ama has GH¢120. She spends $\frac{1}{3}$ of it on food and $\frac{1}{4}$ on transport. How much is left?
- A. GH¢70
 - B. GH¢40
 - C. GH¢110
 - D. GH¢50
- 38.** A rope $7\frac{1}{2}$ m long is cut into pieces each $1\frac{1}{2}$ m long. How many pieces are obtained?
- A. 5
 - B. 6
 - C. 3
 - D. 4
- 39.** $\frac{5}{7}$ of a class of 42 pupils are girls. How many boys are in the class?
- A. 21
 - B. 30
 - C. 7
 - D. 12
- 40.** Express $\frac{2}{3}$ as a decimal, correct to 2 decimal places.
- A. 0.66
 - B. 0.67
 - C. 0.6
 - D. 0.7

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

(a) Review of fractions

Question 2

(a) Addition and subtraction of fractions

Question 3

(a) Multiplication, division and BODMAS

Question 4

(a) Fractions of quantities

Question 5

(a) Word problems with rounding

Question 6

(a) Create and solve

Chapter 16: Ch7

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Express 12 : 18 in its simplest form.
 - A. 3 : 2
 - B. 2 : 3
 - C. 4 : 6
 - D. 12 : 18
2. GH¢120 is shared between Ama and Kofi in the ratio 2 : 3. How much does Ama receive?
 - A. GH¢48
 - B. GH¢72
 - C. GH¢60
 - D. GH¢50
3. In a class, the ratio of boys to girls is 3 : 5. If there are 40 pupils, how many boys are there?
 - A. 12
 - B. 15
 - C. 25
 - D. 24
4. A car travels 180 km in 3 hours. Find its speed.
 - A. 540 km/h
 - B. 90 km/h
 - C. 45 km/h
 - D. 60 km/h
5. If 4 pens cost GH¢12, find the cost of 7 pens.
 - A. GH¢14
 - B. GH¢21
 - C. GH¢24
 - D. GH¢28
6. The total cost t of items is proportional to the number n at GH¢4.50 each ($t = pn$). Find t when $n = 6$.
 - A. GH¢27.00
 - B. GH¢13.50
 - C. GH¢10.50

D. GH¢27.50

- 7.** The graph of two quantities is a straight line through the origin. The relationship is
- A. inversely proportional
 - B. constant
 - C. directly proportional
 - D. zero
- 8.** How much do 8 avocados cost?
- A. GH¢10.50
 - B. GH¢12.00
 - C. GH¢16.00
 - D. GH¢8.00
- 9.** Using the graph, how much do 15 avocados cost?
- A. GH¢24.00
 - B. GH¢15.00
 - C. GH¢20.00
 - D. GH¢22.50
- 10.** The unit rate of the graph (its slope) is
- A. GH¢1.50 per avocado
 - B. GH¢3.00 per avocado
 - C. GH¢2.00 per avocado
 - D. GH¢0.50 per avocado
- 11.** In the simple interest formula $I = PRT/100$, the letter R represents the
- A. time in years
 - B. amount
 - C. principal
 - D. rate of interest per annum
- 12.** Find the simple interest on GH¢4,500 invested at 3% per annum for 3 years.
- A. GH¢4,905
 - B. GH¢405
 - C. GH¢135
 - D. GH¢450
- 13.** The amount at the end of the 3 years in the previous question is
- A. GH¢4,905
 - B. GH¢4,505
 - C. GH¢4,635
 - D. GH¢5,175

- 14.** The amount at the end of the fifth year is
- A. GH¢5,025
 - B. GH¢5,850
 - C. GH¢4,905
 - D. GH¢5,175
- 15.** Find the amount on GH¢2,000 invested at 5% per annum simple interest for 2 years.
- A. GH¢2,500
 - B. GH¢2,100
 - C. GH¢2,200
 - D. GH¢2,020
- 16.** The simple interest on a principal for 4 years at 5% per annum is GH¢300. The principal is
- A. GH¢60
 - B. GH¢1,500
 - C. GH¢3,000
 - D. GH¢1,200
- 17.** The VAT rate in Ghana is 12.5%. The VAT on an item priced GH¢400 (exclusive) is
- A. GH¢450
 - B. GH¢500
 - C. GH¢12.50
 - D. GH¢50
- 18.** A man bought an item at GH¢4,500.00, VAT inclusive, at 12.5%. The basic cost of the item is
- A. GH¢4,500
 - B. GH¢4,000
 - C. GH¢3,937.50
 - D. GH¢4,062.50
- 19.** The VAT paid by the man is
- A. GH¢562.50
 - B. GH¢450
 - C. GH¢625
 - D. GH¢500
- 20.** A car costs GH¢80,500.00. Its new value after a discount of 10% is
- A. GH¢72,450
 - B. GH¢72,500
 - C. GH¢70,450
 - D. GH¢79,695

- 21.** The discount allowed on the car is
- A. GH¢10,050
 - B. GH¢8,500
 - C. GH¢8,050
 - D. GH¢805
- 22.** An agent's commission on the sale of a car is $3\frac{1}{2}\%$. The commission on a car sold for GH¢68,000.00 is
- A. GH¢238
 - B. GH¢2,040
 - C. GH¢3,400
 - D. GH¢2,380
- 23.** A mobile phone originally worth GH¢1,800 depreciates by 5% of its original value in the first year. Its value at the end of the first year is
- A. GH¢1,755
 - B. GH¢1,710
 - C. GH¢90
 - D. GH¢1,620
- 24.** In the second year the phone depreciates by 10% of the original value. Its value at the end of the second year is
- A. GH¢1,710
 - B. GH¢1,530
 - C. GH¢1,620
 - D. GH¢1,638
- 25.** The NHIL inclusive price of a television set is GH¢1,200.00. If the NHIL is charged at 2.5%, the cost of the television (NHIL exclusive) is
- A. GH¢1,170.73
 - B. GH¢1,170.00
 - C. GH¢30.00
 - D. GH¢1,230.00
- 26.** The NHIL charged is
- A. GH¢25.00
 - B. GH¢29.00
 - C. GH¢29.27
 - D. GH¢30.00
- 27.** Kofi Mereku paid an insurance premium of GH¢30,000.00. If the premium is 5% of the value of the house, the insured value is
- A. GH¢60,000
 - B. GH¢150,000
 - C. GH¢1,500,000

D. GH¢600,000

28. Under Act 766, the SSNIT contribution rate of the worker is

- A. 13% of basic salary
- B. 5% of basic salary
- C. 5.5% of basic salary
- D. 18.5% of basic salary

29. Under Act 766, the SSNIT contribution rate of the employer is

- A. 12.5% of basic salary
- B. 17.5% of basic salary
- C. 5.5% of basic salary
- D. 13% of basic salary

30. The total monthly SSNIT contribution under Act 766 is

- A. 18.5% of basic salary
- B. 8.5% of basic salary
- C. 17.5% of basic salary
- D. 18% of basic salary

31. A worker's basic monthly salary is GH¢3,256.50. The worker's SSNIT contribution under Act 766 is

- A. GH¢423.35
- B. GH¢162.83
- C. GH¢179.11
- D. GH¢179.00

32. The employer's SSNIT contribution on the same salary is

- A. GH¢423.34
- B. GH¢423.35
- C. GH¢406.56
- D. GH¢179.11

33. The total SSNIT contribution at the end of every month is

- A. GH¢625.00
- B. GH¢602.46
- C. GH¢602.45
- D. GH¢586.17

34. Under PNDCL 247, the worker's SSNIT contribution rate is

- A. 5.5% of basic salary
- B. 5% of basic salary
- C. 12.5% of basic salary
- D. 13% of basic salary

- 35.** Mr Bediako's monthly SSNIT contribution under PNDCL 247 is GH¢440.54. His employer contributes
- A. GH¢440.54
 - B. GH¢1,221.50
 - C. GH¢1,101.35
 - D. GH¢881.08
- 36.** Mr Bediako's basic salary per month is
- A. GH¢8,810.80
 - B. GH¢3,524.32
 - C. GH¢4,405.40
 - D. GH¢8,810.00
- 37.** If $3a = 4b$, then $a : b$ is
- A. 3 : 4
 - B. 12 : 1
 - C. 1 : 12
 - D. 4 : 3
- 38.** A map has a scale of 1 : 50,000. A road measures 4 cm on the map. The actual length of the road is
- A. 2 km
 - B. 200 m
 - C. 200,000 km
 - D. 20 km
- 39.** 5 workers take 12 days to complete a job. At the same rate, 10 workers will take
- A. 4 days
 - B. 24 days
 - C. 6 days
 - D. 9 days
- 40.** y is directly proportional to x , so $y = kx$. If $y = 12$ when $x = 3$, then $k =$
- A. 4
 - B. 36
 - C. 15
 - D. 9

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Ratio and sharing

Question 2

- (a) Rates and proportion

Question 3

- (a) Simple interest

Question 4

- (a) VAT, discount and commission

Question 5

- (a) SSNIT contributions
(a) 5% of basic salary
(a) 0% of basic salary

Question 13

- (a) 0% of basic salary

Question 12

- (a) 5% of basic salary

Question 18

- (a) 5% of basic salary

Question 17

- (a) 5% of basic salary

Question 6

- (a) Graphing a proportional relationship

Chapter 17: Ch

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. The value of y when $x = 3$ on the line $y = 2x + 1$ is
 - A. 5
 - B. 6
 - C. 8
 - D. 7
2. The value of y when $x = -3$ on the relation $y = -x + 5$ is
 - A. 8
 - B. -2
 - C. 3
 - D. 2
3. The value of y when $x = 4$ on the relation $y = \frac{1}{2}x - 3$ is
 - A. -2
 - B. 1
 - C. -1
 - D. -5
4. The value of y when $x = -2$ on the line $y = 3x - 2$ is
 - A. -8
 - B. -4
 - C. -6
 - D. 4
5. Which ordered pair lies on the line $y = x + 4$?
 - A. $(-4, 1)$
 - B. $(-1, 3)$
 - C. $(0, -4)$
 - D. $(1, 3)$
6. Which point does NOT lie on the line $y = 2x - 1$?
 - A. $(2, 4)$
 - B. $(0, -1)$
 - C. $(1, 1)$
 - D. $(3, 5)$

- 7.** Completing the table for $y = -x + 5$: when $x = 0$, y is
- A. 0
 - B. -5
 - C. 5
 - D. 1
- 8.** Completing the table for $y = \frac{1}{2}x - 3$: when $x = -2$, y is
- A. -1
 - B. -4
 - C. 2
 - D. -2
- 9.** The gradient of the line $y = -x + 4$ is
- A. -4
 - B. 1
 - C. -1
 - D. 4
- 10.** The y-intercept of the line $y = x - 2$ is
- A. 2
 - B. 0
 - C. 1
 - D. -2
- 11.** On the line $y = -x + 4$, when $x = -1$, y is
- A. -5
 - B. -3
 - C. 5
 - D. 3
- 12.** On the line $y = x - 2$, when $x = -1$, y is
- A. 1
 - B. -3
 - C. -1
 - D. 3
- 13.** The coordinates of the intersection of $y = -x + 4$ and $y = x - 2$ are
- A. (1, 3)
 - B. (2, 2)
 - C. (3, 1)
 - D. (-3, 7)

- 14.** The line $y = x - 2$ crosses the x-axis at $x =$
- A. 0
 - B. 2
 - C. 4
 - D. -2
- 15.** The line $y = -x + 4$ meets the y-axis at
- A. (0, 4)
 - B. (1, 4)
 - C. (4, 0)
 - D. (0, -4)
- 16.** The gradient of the line $y = \frac{1}{2}x - 3$ is
- A. -3
 - B. 2
 - C. $\frac{1}{2}$
 - D. $-\frac{1}{2}$
- 17.** Solving $y = -x + 7$ and $y = 2x + 1$ simultaneously gives
- A. (1, 6)
 - B. (2, 5)
 - C. (3, 4)
 - D. (5, 2)
- 18.** Solving $y = x + 4$ and $y = 6 - x$ simultaneously gives
- A. (-1, 5)
 - B. (5, 1)
 - C. (2, 4)
 - D. (1, 5)
- 19.** Solving $y = 2x$ and $y = x + 3$ simultaneously gives
- A. (6, 3)
 - B. (3, 6)
 - C. (3, 5)
 - D. (2, 4)
- 20.** Solving $y = x + 1$ and $y = 2x - 1$ simultaneously gives
- A. (1, 2)
 - B. (2, 1)
 - C. (2, 3)
 - D. (3, 2)

- 21.** The lines $y = 2x + 1$ and $y = 2x - 3$ have
- A. no solution (they are parallel)
 - B. the solution $(1, 3)$
 - C. exactly one solution
 - D. infinitely many solutions
- 22.** The equations $y = x + 2$ and $y = x + 2$ have
- A. no solution
 - B. exactly one solution
 - C. the solution $(0, 2)$ only
 - D. infinitely many solutions
- 23.** The graphs of $y = 2x - 1$ and $y = -x + 5$ intersect at
- A. $(1, 1)$
 - B. $(2, 3)$
 - C. $(0, 5)$
 - D. $(3, 2)$
- 24.** Using the graph, when $x = 4$ the value of y on $y = 2x - 1$ is
- A. 7
 - B. 3
 - C. 8
 - D. 6
- 25.** Solve simultaneously: $x + y = 7$ and $x - y = 3$.
- A. $x = 2, y = 5$
 - B. $x = 5, y = 2$
 - C. $x = 4, y = 3$
 - D. $x = 5, y = 3$
- 26.** Solve simultaneously: $2x + y = 8$ and $x + y = 5$.
- A. $x = 4, y = 0$
 - B. $x = 3, y = 3$
 - C. $x = 2, y = 3$
 - D. $x = 3, y = 2$
- 27.** Solve simultaneously: $3x - y = 7$ and $x + y = 5$.
- A. $x = 1, y = 4$
 - B. $x = 3, y = 2$
 - C. $x = 2, y = 1$
 - D. $x = 3, y = -2$

- 28.** The sum of two numbers is 10 and their difference is 4. The numbers are
- A. 6 and 4
 - B. 8 and 2
 - C. 5 and 5
 - D. 7 and 3
- 29.** A pen costs GH¢2 more than a pencil. Three pens and two pencils together cost GH¢16. The price of a pen is
- A. GH¢4
 - B. GH¢5
 - C. GH¢6
 - D. GH¢3
- 30.** Ama is 3 years older than Kofi. The sum of their ages is 25. Ama's age is
- A. 12
 - B. 13
 - C. 15
 - D. 14
- 31.** Two goats and one sheep cost GH¢700; one goat and two sheep cost GH¢800. The cost of a goat is
- A. GH¢300
 - B. GH¢250
 - C. GH¢200
 - D. GH¢350
- 32.** Solving $y = 3x$ and $y = x + 8$ simultaneously gives
- A. (12, 4)
 - B. (4, 20)
 - C. (8, 24)
 - D. (4, 12)
- 33.** The value of y when $x = 3$ on the relation $y = -x + 5$ is
- A. 2
 - B. 3
 - C. -2
 - D. 8
- 34.** The value of y when $x = 0$ on the relation $y = \frac{1}{2}x - 3$ is
- A. 3
 - B. 0
 - C. -1.5
 - D. -3

- 35.** The relation shown by the table $x: 0, 1, 2, 3; y: 1, 3, 5, 7$ is
- A. $y = 2x + 1$
 - B. $y = 2x - 1$
 - C. $y = 3x + 1$
 - D. $y = x + 1$
- 36.** Which table of values represents $y = x - 2$?
- A. $(-2, 0), (0, 2), (2, 4)$
 - B. $(0, 2), (2, 4), (4, 6)$
 - C. $(0, -2), (2, 0), (4, 2)$
 - D. $(0, 2), (2, 0), (4, -2)$
- 37.** The graph of $y = -x + 5$ is
- A. a line sloping downwards from left to right, crossing the y-axis at 5
 - B. a line sloping upwards from left to right, crossing the y-axis at 5
 - C. a line sloping downwards from left to right, crossing the y-axis at -5
 - D. a line sloping upwards from left to right, crossing the y-axis at -5
- 38.** The line $y = \frac{1}{2}x - 3$ crosses the x-axis at $x =$
- A. -6
 - B. 6
 - C. -3
 - D. 3
- 39.** The graphs of two relations intersect at $(2, 5)$. The solution of the simultaneous equations is
- A. $x = 7$
 - B. $x = 5$ and $y = 2$
 - C. $x = 2$ only
 - D. $x = 2$ and $y = 5$
- 40.** Graphically, the solution of simultaneous linear equations is found at
- A. the y-intercepts of the lines
 - B. the x-intercepts of the lines
 - C. the intersection of the two lines
 - D. the midpoint of the lines

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Tables of values and intersection

Question 2

- (a) Graphing two linear relations

Question 3

- (a) Solving simultaneous equations algebraically

Question 4

- (a) Word problems

Question 5

- (a) Reading the graph (Fig. 1)

Question 6

- (a) Create and solve

Chapter 18: Ch9

Time Allowed: 2 Hours

Total Marks: 100 | Total Marks: 100

SECTION A: OBJECTIVE TEST

Answer ALL questions. Each question carries 1 mark.

1. Make x the subject of the formula $x + a = b$.

A. $x = a - b$
B. $x = b - a$
C. $x = b + a$
D. $x = -b - a$

2. Make x the subject of the formula $2x = y$.

A. $x = y - 2$
B. $x = y + 2$
C. $x = y/2$
D. $x = 2y$

3. Make r the subject of the formula $C = 2\pi r$.

A. $r = C/(2\pi)$
B. $r = C - 2\pi$
C. $r = C/\pi$
D. $r = 2\pi C$

4. Make h the subject of the formula $A = \frac{1}{2}bh$.

A. $h = A/b$
B. $h = A/(2b)$
C. $h = \frac{1}{2}Ab$
D. $h = 2A/b$

5. Make t the subject of the formula $v = u + at$.

A. $t = (v - u)/a$
B. $t = (v - u) \times a$
C. $t = v - u/a$
D. $t = (v + u)/a$

6. Make b the subject of the formula $A = l \times b$.

A. $b = A - l$
B. $b = A/l$
C. $b = A \times l$
D. $b = l/A$

- 7.** Make x the subject of the formula $y = 3x - 2$.
- A. $x = (y - 2)/3$
 - B. $x = 3(y + 2)$
 - C. $x = (y + 2)/3$
 - D. $x = y/3 + 2$
- 8.** Make r the subject of the formula $A = \pi r^2$.
- A. $r = \sqrt{A\pi}$
 - B. $r = \sqrt{A/\pi}$
 - C. $r = 2A/\pi$
 - D. $r = A/\pi$
- 9.** Make l the subject of the formula $P = 2(l + b)$.
- A. $l = P/2 - b$
 - B. $l = 2P - b$
 - C. $l = P - 2b$
 - D. $l = (P - b)/2$
- 10.** Make a the subject of the formula $v^2 = u^2 + 2as$.
- A. $a = (v - u)/(2s)$
 - B. $a = (v^2 - u^2) \times 2s$
 - C. $a = (v^2 - u^2)/(2s)$
 - D. $a = (v^2 + u^2)/(2s)$
- 11.** The area of a rectangle is 24 cm^2 . If the length is 8 cm , the width is (curriculum exemplar)
- A. 16 cm
 - B. 6 cm
 - C. 4 cm
 - D. 3 cm
- 12.** A circle has an area of 154 cm^2 . Taking $\pi = 22/7$, its radius is (curriculum exemplar)
- A. 14 cm
 - B. 7 cm
 - C. 49 cm
 - D. 22 cm
- 13.** The triangle has an area of 54 cm^2 . The value of the height h is (curriculum exemplar)
- A. 4.5 cm
 - B. 108 cm
 - C. 9 cm
 - D. 12 cm

- 14.** The cylinder has a volume of $1,540 \text{ cm}^3$. Taking $\pi = 22/7$, the value of the height h is
- A. 44 cm
 - B. 10 cm
 - C. 22 cm
 - D. 100 cm
- 15.** Find the value of $x^2 + 3x$ when $x = 2$.
- A. 8
 - B. 7
 - C. 12
 - D. 10
- 16.** Evaluate $2a - 3b$ when $a = 4$ and $b = 1$.
- A. 5
 - B. 11
 - C. -1
 - D. 6
- 17.** Find the value of $(x - b)^2 - 3(x - b)$ when $x = 2$ and $b = -5$ (curriculum exemplar).
- A. 49
 - B. 21
 - C. 7
 - D. 28
- 18.** Evaluate $3m^2 - 2m + 1$ when $m = -1$.
- A. 6
 - B. 0
 - C. 2
 - D. 4
- 19.** Evaluate $(p + q)(p - q)$ when $p = 5$ and $q = 3$.
- A. 8
 - B. 16
 - C. 2
 - D. 34
- 20.** Using $s = \frac{1}{2}gt^2$, find s when $g = 10$ and $t = 3$.
- A. 90
 - B. 30
 - C. 15
 - D. 45

- 21.** The volume of the shape is $\frac{1}{3}\pi r^2 h$. The volume when $r = 7$, $h = 21$ and $\pi = \frac{22}{7}$ is (curriculum exemplar)
- A. $1,078 \text{ cm}^3$
 - B. 462 cm^3
 - C. $3,234 \text{ cm}^3$
 - D. 154 cm^3
- 22.** Using $I = \frac{PRT}{100}$, find I when $P = 500$, $R = 4$ and $T = 3$.
- A. 600
 - B. 120
 - C. 6,000
 - D. 60
- 23.** Using $v = u + at$, find v when $u = 5$, $a = 2$ and $t = 4$.
- A. 14
 - B. 13
 - C. 7
 - D. 11
- 24.** Evaluate $a^2 - b^2$ when $a = 7$ and $b = 3$.
- A. 44
 - B. 16
 - C. 4
 - D. 40
- 25.** Evaluate $(2x + y)/3$ when $x = 5$ and $y = 2$.
- A. 3
 - B. $3 \frac{1}{3}$
 - C. 4
 - D. 6
- 26.** Evaluate $\sqrt{a^2 + b^2}$ when $a = 3$ and $b = 4$.
- A. 5
 - B. 25
 - C. 1
 - D. 7
- 27.** Factorise $3x + 4xy$ (curriculum exemplar).
- A. $xy(3 + 4)$
 - B. $x(4 + 3y)$
 - C. $3x(1 + 4y)$
 - D. $x(3 + 4y)$

28. Factorise $12ab + 16b$ (curriculum exemplar).

- A. $4a(3b + 4)$
- B. $4b(3a + 4b)$
- C. $4b(3a + 4)$
- D. $16b(3a + 1)$

29. Factorise $-13xy + 39x$ (curriculum exemplar).

- A. $-13x(y + 3)$
- B. $-13x(y - 3)$
- C. $-13x(3 - y)$
- D. $13x(y - 3)$

30. Factorise $8y - 2y^2$ (curriculum exemplar).

- A. $y(8 - y)$
- B. $2y(4 - 2y)$
- C. $2y(4 - y)$
- D. $2y(4 + y)$

31. Factorise $5x + 10$.

- A. $5(x + 2)$
- B. $5(x + 10)$
- C. $x(5 + 2)$
- D. $5x + 2$

32. Factorise $x^2 + 3x$.

- A. $x^2 + 3$
- B. $3x(x + 1)$
- C. $x(x + 3)$
- D. $(x + 3)$

33. Factorise $6a^2 - 9a$.

- A. $6a(a - 9)$
- B. $3a(2a - 3)$
- C. $3a(2a + 3)$
- D. $9a(a - 3)$

34. Factorise completely $4x^2 - 16$.

- A. $(x + 4)(x - 4)$
- B. $4(x - 2)^2$
- C. $(2x - 4)(x + 2)$
- D. $4(x + 2)(x - 2)$

35. Factorise $ax + ay + bx + by$.

- A. $(a + b)(x + y)$
- B. $(a - b)(x + y)$
- C. $(a + x)(b + y)$
- D. $(a + b)(x - y)$

36. Factorise $2x^2 + 5x + 2$.

- A. $(2x + 1)(x - 2)$
- B. $(2x - 1)(x - 2)$
- C. $(x + 1)(2x + 2)$
- D. $(2x + 1)(x + 2)$

37. Factorise $7p - 7q$.

- A. $7(p - q)$
- B. $7(p + q)$
- C. $7p - q$
- D. $p - 7q$

38. Factorise $x^2 - 9$.

- A. $(x + 9)(x - 1)$
- B. $(x + 3)^2$
- C. $(x + 3)(x - 3)$
- D. $(x - 3)^2$

39. The highest common factor of $6x^2y$ and $9xy^2$ is

- A. $6xy$
- B. $3xy$
- C. xy
- D. $3x^2y^2$

40. Factorise $15m^2n - 10mn^2$.

- A. $5mn(3m + 2n)$
- B. $5mn(3 - 2n)$
- C. $5mn(3m - 2n)$
- D. $10mn(3m - 2n)$

SECTION B: STRUCTURED QUESTIONS

Answer all questions and show all your workings clearly.

Question 1

- (a) Change of subject

Question 2

- (a) Using changed subjects with the figures

Question 3

- (a) Substitution

Question 4

- (a) Factorisation (curriculum exemplars)

Question 5

- (a) Mixed: substitution and factorisation

Question 6

- (a) Create and solve

ANSWER KEY

Ch12**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
36	A	Mathematics	Apply

Answer Distribution:

A: 1 (100%)

Cognitive Level Distribution:

Apply: 1 (100%)

Ch13**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Mathematics	Apply
2	A	Mathematics	Apply
3	A	Mathematics	Apply
4	A	Mathematics	Apply

Answer Distribution:

A: 4 (100%)

Cognitive Level Distribution:

Apply: 4 (100%)

Ch14**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
37	B	Mathematics	Apply

Answer Distribution:

B: 1 (100%)

Cognitive Level Distribution:

Apply: 1 (100%)

Ch15**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
10	A	Mathematics	Apply
11	A	Mathematics	Apply
12	A	Mathematics	Apply
24	A	Mathematics	Apply

Answer Distribution:

A: 4 (100%)

Cognitive Level Distribution:

Apply: 4 (100%)

Ch16**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
4	A	Mathematics	Apply
18	A	Mathematics	Apply

Answer Distribution:

A: 2 (100%)

Cognitive Level Distribution:

Apply: 2 (100%)

Ch17**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
2	A	Mathematics	Apply
4	A	Mathematics	Apply
33	A	Mathematics	Apply

Answer Distribution:

A: 3 (100%)

Cognitive Level Distribution:

Apply: 3 (100%)

Ch18**ANSWER KEY****SECTION A: OBJECTIVE TEST**

Total Marks: 40

Q	Ans	Curriculum Area	Cognitive Level
1	A	Mathematics	Apply
2	A	Mathematics	Apply

Answer Distribution:

A: 2 (100%)

Cognitive Level Distribution:

Apply: 2 (100%)